

1 Physical Primitives of Mind: Modeling
2 Consciousness via Symmetries and Landauer
3 Erasure

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6 **Abstract**

7 We present a mathematical synthesis and formal Lean 4 verifica-
8 tion demonstrating how cognitive architectures can emerge as attrac-
9 tor states from established physical primitives, without introducing
10 new fundamental physics. Starting from the finiteness of a localized
11 system’s phase space, we construct a framework for predictive process-
12 ing; the spacetime symmetries are the setting rather than a premise,
13 and no theorem consumes them. Spontaneous symmetry breaking gener-
14 ates necessary topological boundaries, and finite phase space sub-
15 jects these boundaries to Landauer’s thermodynamic erasure. Agency

16 arises when systems evolve along paths of least action, minimizing en-
 17 tropy production via internal structural resonance with external per-
 18 turbations. Scaling this dynamic to macroscopic levels aligns with
 19 emerging paradigms in neuroscience—specifically classical electrody-
 20 namic fields (ephaptic coupling)—and Kuramoto phase synchroniza-
 21 tion, culminating in a topological fixed point: a unified “Self.” We
 22 state the claim narrowly, because the subject matter invites a wider
 23 reading than the work supports: no new physics is derived here. Every
 24 physical result the chain invokes is established—Picard–Lindelöf, Kib-
 25 ble’s classification of topological defects, Landauer’s erasure bound,
 26 Gibbs’ inequality, Still’s bound on dissipation by nonpredictive in-
 27 formation, Barbalat’s lemma, Sakaguchi’s threshold, Banach’s fixed
 28 point. What is new is that they are stated in one formal language,
 29 their consistency mechanically checked, and the points where the chain
 30 relies on stipulation rather than derivation located exactly. The chain
 31 itself composes to a single theorem, whose eight unproved steps are its
 32 arguments rather than sentences about it, and which are jointly satis-
 33 fiable rather than jointly empty. The development declares no axioms
 34 and every structure carrying physical content is inhabited by a witness;
 35 the synchronization step is proved from a dynamics for initial data in-
 36 side an explicit basin and assumed outside it, and the identification of
 37 a global section of the probability sheaf with the unity of experience
 38 is a stipulation, marked as such throughout rather than absorbed into
 39 the derivation. The step from a coarse-grained coupling to the brain’s

40 endogenous electromagnetic field is likewise an empirical commitment
41 and not a deduction; we separate it from the coarse-graining theorem
42 that precedes it, weaken it to a modulatory-but-integrative role, and
43 give four conditions that would falsify it. The framework is nonethe-
44 less falsifiable in one place that is its own and no rival's: recovery
45 from sudden awakening should follow a delayed sigmoid rather than
46 an exponential, because the variable gating it is astrocytic volume on
47 a timescale of seconds to minutes rather than cortical phase realign-
48 ment on a timescale of milliseconds. As a corollary, we show why
49 rigid von Neumann architectures (e.g., GPUs) are thermodynamically
50 disadvantaged or structurally constrained from reaching these specific
51 continuous thermodynamic attractors.

52 **1 Introduction**

53 The pursuit of a physical theory of consciousness typically involves search-
54 ing for the biological correlates of mind or treating subjective experience as
55 an abstract computational algorithm. This paper asks a more fundamental
56 question: If we assume only basic physical primitives—continuous symme-
57 tries (the Poincaré group), geometry, and thermodynamics—is the emergence
58 of a unified, perceiving entity a natural physical attractor?

59 We propose a framework where the structural properties of consciousness
60 are derived as physical attractor states. Establishing a deductive chain from
61 basic principles, we demonstrate how continuous symmetries break to form

62 boundaries, how those boundaries dissipate free energy as non-equilibrium
 63 steady states, and how their evolution along paths of least action results
 64 in physical resonance with their environment. This eventually scales into
 65 the macroscopic electromagnetic unity proposed by emerging field theories
 66 of mind. To avoid critiques of “math theatre,” we delineate between literal
 67 physical mechanisms (e.g., thermodynamic erasure via Landauer’s principle,
 68 boundaries from spontaneous symmetry breaking) and theoretical mappings,
 69 utilizing Lean 4 formalization as our mathematical anchor. We adopt a per-
 70 spective aligned with Russellian Monism: objective measurements of macro-
 71 scopic electromagnetic fields constitute the third-person “outside,” while the
 72 system’s internal physical deformation to reach structural resonance consti-
 73 tutes the first-person intrinsic experience.

74 **What is new here, and what is not**

75 A framework paper about consciousness invites a particular suspicion, and it
 76 is better answered at the start than at the end. **No new physics is derived**
 77 **here.** Every physical result the chain invokes is established, and most of
 78 it is old: Picard–Lindelöf for the well-posedness of the coupled dynamics
 79 (Coddington & Levinson, 1955); Kibble’s classification of topological defects
 80 by the homotopy of the vacuum manifold (Kibble, 1976); Landauer’s bound
 81 on the heat cost of erasure (Landauer, 1961); Gibbs’ inequality for the relative
 82 entropy (Cover & Thomas, 2006); Barbalat’s lemma for the asymptotics of a
 83 dissipating trajectory (Barbalat, 1959); the threshold $K_c = 2D$ for the noisy

84 mean-field Kuramoto model (Sakaguchi, 1988); the bound of Still, Sivak,
 85 Bell and Crooks relating dissipation to nonpredictive information (Still et
 86 al., 2012); the identity of Kawai, Parrondo and Van den Broeck equating
 87 dissipated work with the relative entropy between forward and time-reversed
 88 ensembles (Kawai et al., 2007); and Banach’s fixed-point theorem (Banach,
 89 1922). A physicist reading this paper will recognize all of it and would
 90 establish the point in three paragraphs if we did not. Conceding it costs
 91 nothing.

92 What is new is of a different kind, and it is a smaller claim than a theory
 93 of consciousness usually makes. These results have been stated in one formal
 94 language, their mutual consistency has been mechanically checked, and the
 95 places where the chain relies on stipulation rather than derivation have been
 96 located exactly. The value of a machine-checked chain is not that any link
 97 is novel; it is that the claim being made *about* the chain is itself checkable,
 98 and we state it in the form a reader can check rather than in the form that
 99 asks to be believed. Four things. The development declares **no axioms**:
 100 **#print axioms** on any result reports only **propext**, **Classical.choice** and
 101 **Quot.sound**. Every physical postulate is a field of a class that each model
 102 discharges for itself rather than a global assertion, and every such class is
 103 inhabited by an explicit witness, so no theorem is vacuous for want of a model.
 104 The chain composes to a *single* theorem — **chain**, in **Chain.lean** — which
 105 derives the Self from the finiteness of the phase space and takes each unproved
 106 step as one of **eight named hypotheses**: four formalization gaps, two

107 modelling assumptions and two physical commitments, each an implication
108 between two links rather than an opaque symbol, so that deleting one fails
109 the build instead of softening a sentence. And the eight hold *simultaneously*
110 on one substrate (`chain_nonvacuous`), which rules out the failure in which
111 a composition proves its conclusion only because its premises cannot all be
112 met at once. The seams — where physics stops and interpretation begins —
113 are thereby fixed in the source rather than negotiable in the prose, and their
114 number is a number rather than an impression.

115 Composing the chain also refuted two of its own arrows, and Figure 1
116 draws the corrected picture rather than repairing it quietly. The capacity
117 bound of the first link has no hypotheses at all, so an implication from it to
118 the existence of a boundary is equivalent to assuming the boundary outright:
119 those two links are independent starting points and not consecutive steps.
120 What finiteness does imply directly is the heat cost, skipping the boundary
121 entirely. Neither of the two arrows that are theorems is one the chain’s
122 usual narration draws. Section 2.1 reports what the check cost: three of five
123 original axioms each proved **False**, from one shared root cause, and none of
124 the three had been caught by reading.

125 The framework is not thereby unfalsifiable, and one of its predictions is
126 its own rather than a rival’s. Section 11 argues that recovery from sudden
127 awakening should follow a *delayed sigmoid* rather than an exponential, be-
128 cause the variable gating recovery is astrocytic volume and cerebrospinal-fluid
129 clearance, on a timescale of seconds to minutes, while the cortical rhythms

130 whose realignment is usually taken to constitute the recovery move in mil-
131 liseconds. That is a claim about functional form; it is fittable against data of
132 a kind already collected; and a purely electrical account of the same recovery
133 predicts a different form.

134 2 Formalization in Lean 4

135 A core contribution of this work is the verification of our deductive chain
136 using the Lean 4 interactive theorem prover. This approach builds on recent
137 efforts to formalize consciousness and brain semantics, such as the 'OnlyOne'
138 system (Scherf, 2026) and sheaf-theoretic models of neural manifolds (Inoué,
139 2026). It is necessary to explicitly distinguish which claims have been math-
140 ematically derived as theorems within the proof assistant, and which remain
141 irreducible physical postulates. Figure 1 shows the chain with the status of
142 every node *and of every arrow* marked, and Table 1 states in seven rows
143 what each link assumes. The complete claim-by-claim map to Lean identi-
144 fiers, which runs to several pages, is Table S1 in the supplement; the main
145 text does not depend on reading it.

146 2.1 Soundness of the formalization

147 Formalization is only as trustworthy as the axioms it rests on, and an axiom
148 that is inconsistent silently makes every theorem downstream of it vacuous
149 while leaving the build green. We report a case of exactly this, found by

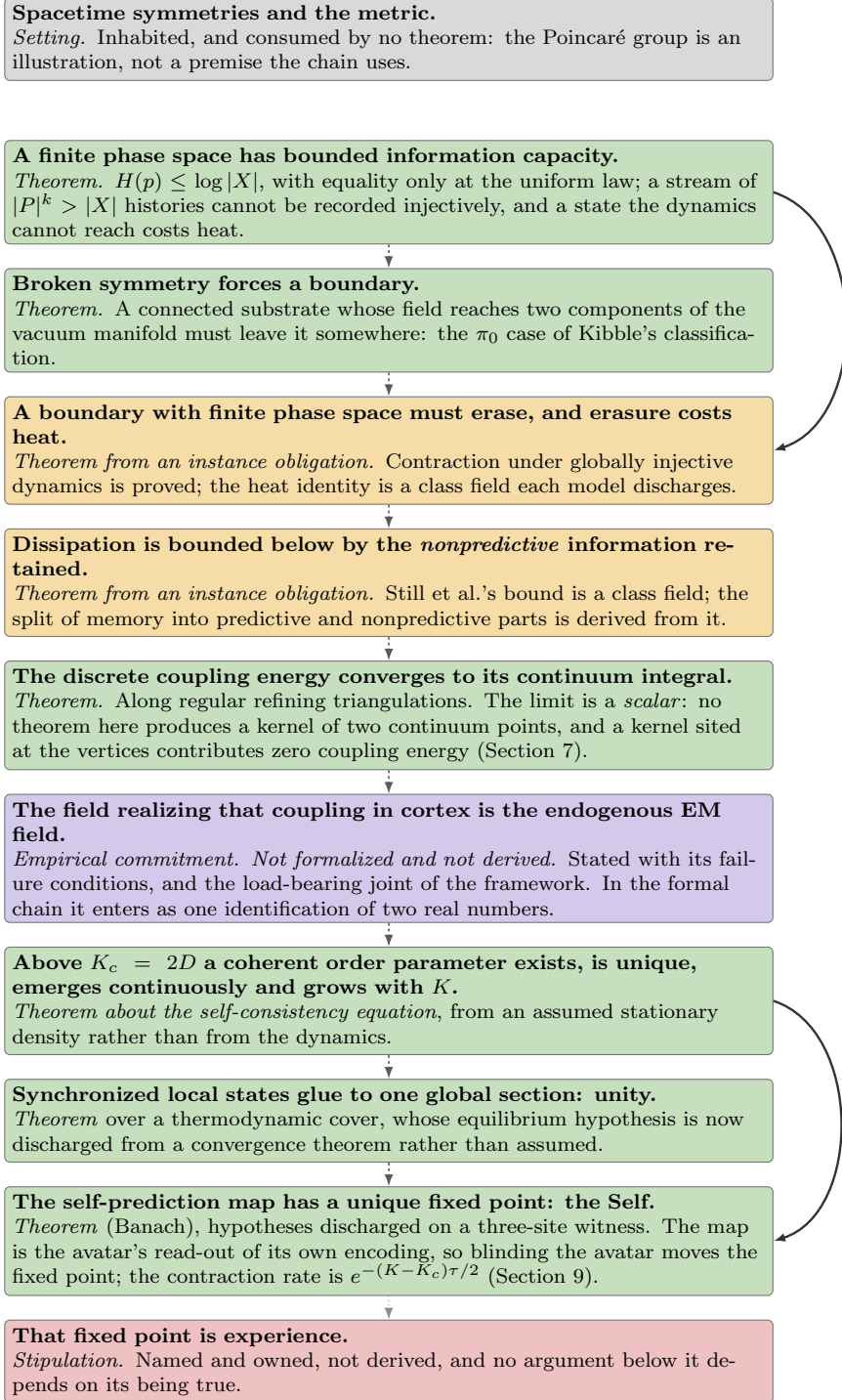


Figure 1: The deductive chain, with⁸ the status of each node and of each arrow. *Nodes.* ■ theorem; ■ theorem resting on a class field that each model must discharge; ■ empirical commitment, neither formalized nor derived; ■ stipulation; ■ setting, consumed by no theorem. *Arrows.* A **solid** arrow is a Lean theorem. A **dashed** arrow is a named hypothesis of chain (Chain.lean): an implication between the two node statements which the development does not prove and which is therefore an argument the theorem

Claim	Status	What is assumed
A finite phase space has bounded information capacity	Theorem	Finiteness of the state space, and nothing else — no symmetry group, no metric. $H \leq \log X $, with equality only at the uniform law. On an infinite state space the step from “unreachable state” to “erasure” fails, and the failure is exhibited rather than assumed away.
Broken symmetry forces a boundary	Theorem	Substrate connected; vacuum manifold disconnected. No hypothesis mentions energy, a potential, or a symmetry.
Erasure costs heat	Theorem + class field	Global system–bath dynamics injective (Liouville). The heat identity <code>heat_eq</code> is discharged per instance, not asserted for all models.
Dissipation is bounded below by nonpredictive information	Theorem + class field	Still et al.’s bound is discharged per instance; the environment evolves by a Markov kernel the system does not act on.
The discrete coupling energy converges to its continuum integral; the field carrying that coupling is the cortical EM field	Theorem; then empirical	Convergence holds along regular refining triangulations, for the energy functional and not the dynamics. The limit is a scalar: no theorem produces a kernel of two continuum points, and a kernel sited at the triangulation’s vertices contributes zero coupling energy on an atomless substrate. The physical identification is a commitment with stated failure conditions (Section 7).
Above $K_c = 2D$ a unique coherent state exists, and synchronized patches glue into a unique global section	Theorem 9	Von Mises stationary density assumed rather than derived from the Fokker–Planck operator; couplings positive and symmetric, so unfrustrated. Gluing needs a cover at the potential minimum whose synchronized patches agree on overlaps; the first is derived from a convergent trajectory, the second is an instance obligation throughout.
The coherent order parameter carries a unique reflexive fixed point:	Theorem	The self-prediction map is the avatar’s read-out of its own encoding, and its Lipschitz rate

auditing our own development, because the failure mode is easy to reproduce and we have not seen it discussed in the applied-formalization literature.

The failure has one shape, and naming it is more useful than cataloguing its instances: *an axiom that constrains a symbol it does not itself bind*. Such a declaration is not a postulate about one system. It is a constraint on every model of the class the symbol belongs to, including models built to violate it, and a class that has such a model is thereby inconsistent. Three concrete cases, each of which proves **False** on its own:

- `spontaneous_symmetry_breaking_pointwise_min` took the potential-energy functional as an *implicit* argument occurring only in its hypothesis `PotentialEnergy phi = V v0`. Instantiating that functional with the constant map $\lambda_. V(v_0)$ discharges the hypothesis by `rfl` and yields the conclusion for an arbitrary field. With $V(v) = v^2$, $v_0 = 0$ and $\phi \equiv 1$ this proves $1 = 0$.
- `landauer_heat_eq` equated the free class field `heat_dissipation` with a bath-entropy difference *for every instance* of the class. An instance with zero heat dissipation and a bath that gains one bit gives $0 = \log 2$.
- `kl_bound_axiom` asserted $\sigma \geq D_{KL}(P||Q)/\Delta t$ for *all* P and Q , while σ is a single real fixed by a class field. Since D_{KL} is unbounded above, a point mass against a uniform distribution refutes it.

Two further declarations of the same shape are worth recording because their refutation is different in kind. `phase_invariant_periodic` and

172 `sync_to_section_eq`, as global assertions pinning free fields of `LocalSectionSynchronization`,
173 are refutable as soon as the probability presheaf admits two distinct global
174 sections — so the refutation needs no arithmetic at all, only that the sheaf is
175 not trivial. Neither is a postulate in the development: the periodicity condi-
176 tion is an obligation of the constructor that builds a cover from an invariant
177 measure, and `sync_to_section_eq` is a theorem about the global section the
178 sheaf condition produces (Derivation 5).

179 The rule that prevents it is uniform, and we state it as the rule we follow:
180 *a physical postulate that mentions a class field must be a field of that class,*
181 *never a standalone axiom quantified over all instances.* A postulate is then
182 an obligation each model discharges, not a global claim about every model.
183 Under that rule the development declares no axioms at all. Two of the
184 statements above are not postulates in the first place — pointwise vacuum
185 minimization is proved from $\int f = 0 \wedge f \geq 0 \Rightarrow f = 0$ a.e., and Gibbs’
186 inequality is Mathlib’s — and the genuinely physical ones, Landauer’s heat
187 identity and the two information bounds, are instance obligations.

188 This makes non-vacuity the live question, since an uninhabitable class
189 makes its theorems vacuous just as surely as an inconsistent axiom did.
190 `Examples.lean` therefore constructs witnesses: a one-bit erasure system with
191 a genuine bath that discharges `heat_eq` and reproduces $\Delta Q = k_B T \ln 2$; a
192 perfectly resonant system discharging `kl_bound`; and a scalar field on a one-
193 point spacetime discharging `ActionPrinciples`. Derivation 5’s gluing struc-
194 tures are the most demanding case and are witnessed too: `Examples.lean`

195 builds a `ThermodynamicCover` over a three-site substrate carrying the dis-
 196 crete topology and its Borel σ -algebra, covered by two *overlapping* open
 197 patches whose intersection is a single shared site. The local sections are
 198 germ families of a phase-indexed finite measure $\mu_\theta = (1 + \cos \theta)^+ \delta_{\text{mid}}$ pushed
 199 into the sheafification of the probability presheaf; the family is provably
 200 non-constant in θ (`phaseMeasure_not_const`), so the periodicity obligation
 201 of the constructor `LocalSectionSynchronization.ofInvariantMeasure` is
 202 discharged by an argument about `cos` rather than satisfied trivially, and
 203 `thermodynamic_equilibrium` is *derived* from `phase_locked_minimizes_potential`
 204 rather than assumed. `global_section_from_thermodynamics` then fires on
 205 this instance and glues the two patch-local sections into a unique global one.
 206 Two further covers answer the follow-up question of how non-degenerate such
 207 a witness can be made. `Examples.lean` §13 has a negative half worth record-
 208 ing: the phase field can be made non-constant as a function, but not as a
 209 physical state, because `thermodynamic_equilibrium` forces every instance
 210 to a phase-locked configuration. `Examples.lean` §14 is built the other way
 211 round, from two patches with independently chosen mass profiles that meet
 212 only on the site they share, and the section the gluing returns is neither of
 213 them. That is the same kind of result as `contracting_implies_const` in
 214 Derivation 6 — a proof that the witness is as strong as the class permits,
 215 rather than a witness that could be improved by trying harder.

216 What the witnesses establish is that the classes are inhabitable, so the
 217 theorems about them are not vacuous. What they do not establish is that

218 a cortex is such an instance. In the cover above the phase field is already
 219 constant — every patch locked at $\theta = 0$ — which is precisely what makes
 220 the equilibrium obligation dischargeable; getting a physical system into that
 221 configuration is the content of the informal dynamical argument, not of the
 222 Lean development.

223 The same audit turned up a defect of a different kind, worth record-
 224 ing because it is not an axiom problem and would not have been caught
 225 by `#print axioms. mesh_refinement_convergence` — the claim that dis-
 226 crete edge weights converge to the continuous energy as the mesh is refined
 227 — sat in a “Conjectures” section as an unproved `def ...: Prop`. It was
 228 not merely unproved but false. Nothing required the edge regions to cover
 229 the manifold, and $\text{diam } \emptyset = 0$, so the triangulation whose edge regions are
 230 all empty satisfied the mesh hypothesis while contributing nothing to the
 231 sum, forcing $|0 - \int f| < \varepsilon$ for every $\varepsilon > 0$. Its binder was dead as well: it
 232 quantified over a triangulation and then resolved the vertex type by instance
 233 search rather than through the bound variable, so the variable was never
 234 used. `Phase2_MeshConvergence.lean` replaces it with a theorem. Conver-
 235 gence is now stated along a *sequence* of triangulations of a named region
 236 S , each required to be regular — edge regions measurable, without self-
 237 loops, disjoint across distinct edges, covering S , and anchored to the sim-
 238 plicial data in the sense that a vertex lies in the closure of every region
 239 of an edge at it. Closure rather than membership is forced: cells are dis-
 240 joint, so a shared vertex belongs to at most one of them. The manuscript’s

241 discrete energy $\frac{1}{2} \sum_u \sum_v w(u, v) f(u)$, which samples only the first endpoint
 242 and is therefore not symmetric, is shown to equal exactly the midpoint rule
 243 $\sum_{u < v} w(u, v) \frac{1}{2} (f(u) + f(v))$ over unordered edges; without that identifica-
 244 tion, reading the double sum as a Riemann sum would be a category error.
 245 At fixed mesh the error is at most $\varepsilon \mu(S)$ with ε a modulus of continuity of
 246 the integrand, and convergence follows for uniformly continuous integrands
 247 as the mesh shrinks. The regularity conditions pull against each other, so
 248 `Examples.lean` §6 discharges them with the uniform partition of $[0, 1)$ into
 249 half-open cells and checks that the approximation genuinely moves $\frac{1}{2}$ at
 250 $N = 1$ and $\frac{1}{4}$ at $N = 2$ for the tent function $x \mapsto |x - \frac{1}{2}|$ — rather than
 251 converging trivially. What is proved is convergence, not the $O(1/N^2)$ rate
 252 the accompanying simulation measures; that would need a midpoint error
 253 term with a second derivative, which we have not developed.

254 A third defect was one of bookkeeping across files, and it is the kind that
 255 survives any number of green builds because each half is individually cor-
 256 rect. The development carried *two* Kuramoto potentials. Lyapunov descent
 257 along the dynamics, $\dot{V} = -\sum_i \dot{\theta}_i^2$, was proved for the full potential $V(\theta) =$
 258 $-\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_i - \theta_j) - \sum_i \omega_i \theta_i$, while the characterization of the minimum
 259 — phase-locked in both directions — and everything downstream of it, in-
 260 cluding Derivation 5’s gluing, concerned the coupling term alone. Nothing
 261 chained them, and the gap was not cosmetic: `kuramoto_potential_unbounded_below`
 262 now proves that as soon as one $\omega_i \neq 0$ the full potential is unbounded below
 263 — walk θ out along ω and the bounded cosine sum cannot compensate — so

264 it has no minimum for phase-locking to attain, and the sentence “the phase-
 265 locked state minimizes the Lyapunov potential” was, read literally, false of
 266 the functional the descent theorem was about. `Phase4_RotatingFrame.lean`
 267 closes this by formalizing the standard reduction $\theta_i \mapsto \theta_i - \Omega t$. For identical
 268 natural frequencies $\omega \equiv \Omega$, a trajectory maps to a trajectory of the zero-
 269 frequency system — which is exactly the system Derivation 5 already worked
 270 with — the two potentials coincide there, and the descent result transports
 271 onto the coupling potential whose minimizers are the phase-locked states.
 272 Phase differences, phase-locking and the magnitude of the order parameter
 273 are all proved invariant under the change of frame, so nothing observable
 274 is lost in the reduction. `Examples.lean` §7 runs the chain end to end on
 275 two oscillators with a common frequency, discharging every hypothesis. The
 276 restriction is real and we state it plainly: the reduction is exact only for
 277 identical frequencies. A genuine spread of natural frequencies leaves residual
 278 detunings in the reduced system, phase-locking then requires coupling above
 279 a critical K_c , and that K_c — Kuramoto’s threshold for a spread of frequen-
 280 cies, not the noise threshold $2D$ of Section 10 — is nowhere in our Lean
 281 development. It is studied numerically, in Figure 5, as Table S1 records.

282 A fourth defect was dimensional, and it concerned the one row of Table S1
 283 that is still numerical rather than proved. `Phase8_ContinuousField.lean`
 284 defines the critical coupling $K_c = 2D$ and a predicate for being above it.
 285 That predicate compared $\int_x \int_y K(x, y)$ against $2D$ — but a double integral
 286 against an unnormalized measure carries the substrate’s mass twice over, so

287 a constant kernel K on a substrate of total mass m contributed Km^2 . The
 288 comparison therefore depended on how large the substrate was rather than
 289 on how strongly it was coupled, and any kernel whatever could be pushed
 290 above threshold by inflating m . `Examples.lean` §9 exhibits the failure con-
 291 cretely: on a two-site substrate carrying counting measure, a system coupled
 292 at $K = 1/2$ against a threshold of $K_c = 1$ — a factor of two *below* threshold
 293 — has an unnormalized double integral of $2 > 1$, and would satisfy any pred-
 294 icate stated with that integral. The fix is to require the substrate’s measure
 295 to be a probability measure, which is the normalization in which the mean-
 296 field derivation of $K_c = 2D$ is carried out; `mean_field_coupling_const`
 297 then gives that a constant kernel has mean-field strength exactly K , and
 298 `exhibits_phase_transition_const_iff` reduces the predicate to $K > 2D$.
 299 We flag what this does and does not do. It makes the predicate say something
 300 whose truth depends only on the physics, and §9 witnesses it on both sides of
 301 the threshold. It does not by itself prove $K_c = 2D$: the content of that value
 302 is the bifurcation of the self-consistency equation $r = I_1(Kr/D)/I_0(Kr/D)$
 303 at $K = 2D$, which is a separate question from whether the predicate is di-
 304 mensionally sound. That bifurcation is proved below, in both directions,
 305 from an assumed stationary density; deriving the density itself needs the
 306 Fokker–Planck operator and is not done.

307 We draw no grand moral from this beyond the obvious one. The value
 308 of a proof assistant here was not that it certified our argument, but that it
 309 made a specific, checkable claim about what had been assumed — and that

310 claim turned out to be false in a way no amount of reading the prose would
311 have revealed.

312 **3 Axiom 1: Symmetries as Setting, and the** 313 **Capacity of a Finite Phase Space**

314 Our starting premise relies on fundamental geometry. In our Lean formaliza-
315 tion (`Phase1Primitives.lean`), continuous spacetime symmetries (e.g., the
316 Poincaré group) act upon field states, and the invariance of the system’s total
317 energy under that action is recorded by the class `SymmetryInvariantAction`.
318 Physically, it is such an invariance that Noether’s theorem converts into the
319 conservation of energy and momentum. The scope of the *formal* claim is nar-
320 rower and we state it plainly: the invariance is a class field, i.e. an assumption
321 a model must discharge, and no conserved quantity is ever constructed from
322 it. There is no Noether theorem in our Lean, and nothing downstream of this
323 section depends on one. The class is now inhabited — `Examples.lean` §11
324 exhibits a double-well potential $V(v) = (v^2 - 1)^2$ on a two-point spacetime
325 with the $\mathbb{Z}_2$ action $v \mapsto \pm v$, discharges the invariance because $(\pm v)^2 = v^2$,
326 and proves the vacuum manifold $\{-1, 1\}$ degenerate with the two minima ex-
327 changed by the symmetry, so an energy-minimising field and its image have
328 equal energy and differ. That is symmetry breaking exhibited rather than
329 assumed; it is still not a conserved quantity, and $\mathbb{Z}_2$ being discrete there is
330 no one-parameter family to differentiate along in the first place. Simultane-

ously, discrete permutation symmetries dictate the statistical mechanics of any physical system, defining a strictly finite volume of phase space.¹

Because phase space is finite, any localized physical system possesses a mathematical limit on the amount of information (distinguishable states) it can embody at any given moment. That sentence is the section’s contribution to the chain, and three theorems carry it, in `Phase1_PhaseSpaceCapacity.lean`, whose only hypothesis about the system is that its state space is finite: no symmetry group, no metric, no action functional appears anywhere in them.

The first is the capacity bound itself. A probability distribution p on a finite state space X satisfies $H(p) = -\sum_x p(x) \log p(x) \leq \log |X|$ (`shannon_entropy_le_log_card`), with equality if and only if p is uniform (`shannon_entropy_eq_log_card_iff`). The “only if” half is the part worth stating: it says $\log |X|$ is the exact capacity rather than a convenient over-estimate, so a system holding any distribution other than the flat one over its states holds strictly less than its size suggests. The proof is Gibbs’ inequality against the uniform law — $\log x \leq x - 1$ applied at $x = 1/(|X|p(x))$, with the $p(x) = 0$ case carrying the slack that makes the equality condition sharp — and it is

¹While the Poincaré group and discrete permutations serve as accessible physical examples, the underlying Lean formalization does not hardcode specific physics. It defines the system at a more primitive mathematical level, relying entirely on generic Pseudo-Riemannian manifolds (for spacetime) and finite measure theory (for statistical mechanics). What the formalization supplies is the vocabulary in which symmetries, conserved quantities and phase-space constraints can be *stated*; it does not derive them. All three of the classes involved are now inhabited in `Examples.lean` — the $\mathbb{Z}_2$ action and its invariance in §11, the real line with the Euclidean metric in §12 — but the instances are witnesses of inhabitability, not of physics: no theorem in the development consumes a symmetry group or a metric, and that half of the premise is setting rather than content. The other half of the premise is not setting, and the paragraphs below are what it does.

348 elementary. What was missing here was the statement, not the difficulty.

349 The second says the limit binds. Let the boundary absorb a stream of
350 perturbations drawn from an alphabet P ; a stream of length k has $|P|^k$
351 possible histories. Once $|X| < |P|^k$, the map from histories to final states
352 cannot be injective (`absorb_not_injective`), so two distinct histories leave
353 the system in the same state, and the entropy of the source strictly exceeds
354 anything the system can hold (`source_entropy_exceeds_capacity`). The
355 update map is arbitrary and is allowed to be reversible at every individual
356 step; the collision is forced by counting alone, which is the sense in which the
357 premise, and not the model of the dynamics, is doing the work.

358 The third is what Derivation 2 consumes, and it is the one place where
359 finiteness does something no other hypothesis could. On a finite state space,
360 a map that fails to reach every state is not injective (`is_erasure_of_not_`
361 `surjective`) — which is exactly the erasure condition Landauer’s bound re-
362 quires — so any update of a finite phase space leaving some state unreachable
363 dissipates strictly positive heat (`finite_phase_space_dissipates`). On an
364 infinite phase space the step simply fails: the successor map on $\mathbb{N}$ misses 0 and
365 is injective. `Examples.lean` §19 records that failure next to the witnesses, so
366 the finiteness hypothesis is fenced rather than assumed to be load-bearing,
367 and runs all three results on a single bit: the fair coin attains $\log 2$, a $3/4$ -
368 $1/4$ bit falls strictly short, two binary perturbations already outrun a one-bit
369 memory, and §1’s one-bit thermodynamic system pays heat for a state it
370 cannot reach.

371 None of this derives the finiteness of phase space. Finiteness is assumed,
372 as it is in any statistical-mechanical treatment, and the Poincaré group re-
373 mains an illustration in prose. What has been removed is the gap between
374 assuming the premise and using it.

375 4 Derivation 1: Symmetry Breaking and the 376 Inevitability of Boundaries

377 A perfectly symmetric universe is the uniform vacuum; no distinct entities
378 exist. For any dynamic process to occur, the continuous symmetries of the
379 background field must undergo spontaneous symmetry breaking.

380 When a field drops to a lower, asymmetric energy state, topology dic-
381 tates the natural creation of defects (such as domain walls or monopoles).
382 This constraint necessitates the creation of a physical boundary—a localized
383 topological distinction between an “inside” and an “outside.” While such
384 a defect (e.g., a crack in a crystal lattice) is necessary to define a separate
385 entity, it is not sufficient for agency. The boundary is relational, defined by
386 the twisted geometry of the field surrounding it, and must be coupled with
387 non-equilibrium thermodynamics to exhibit active behaviors.

388 That first sentence is a claim about topology, and it needs a different
389 argument from the energy claim beside it. The energy half — an energy-
390 minimizing field sits in the vacuum manifold almost everywhere (`pointwise_`
391 `vacuum_of_global_min`) — together with the observation that a boundary

392 defect *obstructs* extension over a contractible interior, produces no defect:
 393 both are statements about a field already given. What produces one is
 394 `exists_notMem_of_connectedComponentIn`: if the substrate is connected
 395 and a continuous field takes values in two *different connected components*
 396 of the vacuum manifold M , then somewhere the field is not in M at all.
 397 That excluded region is the wall. This is the π_0 case of Kibble’s classifica-
 398 tion of topological defects by the homotopy of the vacuum manifold (Kibble,
 399 1976), and the proof is elementary in exactly the way it is elementary in the
 400 physics literature: the continuous image of a connected space is connected, a
 401 connected subset of M lies inside one component of M , and the field’s range
 402 would be such a subset if it never left M . No hypothesis mentions energy, a
 403 potential, or a symmetry; connectedness of the substrate and disconnected-
 404 ness of the vacuum are the whole of it.

405 We formalize the π_0 case and stop there. The higher cases — $\pi_1 \neq 0$
 406 forcing vortex lines, $\pi_2 \neq 0$ forcing monopoles — are the rest of Kibble’s
 407 classification and Mathlib’s homotopy-group library would support at least
 408 the next one; they are available and not taken, because the argument this
 409 paper runs uses domain walls and nothing else. The witness is the double
 410 well of `Examples.lean` §11.1, where the vacuum manifold is *computed* to be
 411 exactly $\{-1, 1\}$ (`wellVacuum_eq`) rather than merely known to contain both
 412 minima, and its disconnection is proved in the form the theorem consumes:
 413 no preconnected subset of it contains both, since a preconnected subset of $\mathbb{R}$
 414 is an interval and an interval spanning -1 and 1 contains 0 , which is the top

415 of the barrier. `wellV_domain_wall` then quantifies over *every* continuous field
 416 on *any* connected substrate that reaches both minima, with no formula for
 417 the field anywhere in the statement. That the class is non-empty is checked
 418 separately, because a theorem quantified over all fields is worthless if none
 419 satisfies its hypotheses: the clipped identity on $\mathbb{R}$ does, and the wall it is
 420 forced to have sits at the origin (`kink_zero_notMem`).

421 **5 Derivation 2: The Thermodynamic Con-** 422 **straint**

423 This newly formed topological boundary is subjected to gauge perturbations
 424 from the “outside” environment. Because the internal phase space is finite
 425 (Axiom 1), it cannot absorb an infinite sequence of incoming perturbations
 426 and must overwrite old internal states.

427 By Landauer’s Principle (Landauer, 1961), the erasure of physical in-
 428 formation irreversibly decreases internal entropy, mandating that the sys-
 429 tem exhaust heat into the environment. Philosophers of physics (such as
 430 John Norton (Norton, 2005) and Orly Shenker (Shenker, 2000)) have cri-
 431 tiqued Landauer’s principle, arguing that physicists conflate *logical* erasure
 432 (a many-to-one mapping in abstract computational states) with *thermody-*
 433 *namic* erasure (a many-to-one mapping in physical phase space). They argue
 434 that logical erasure could conceptually be implemented by an injective phys-
 435 ical process without dissipating heat.

436 Our Lean 4 formalization (`Phase3.CombinatorialThermodynamics.lean`)
 437 addresses this critique by defining erasure as a non-injective mapping on the
 438 *physical* state space ($\neg \text{Function.Injective } t$). We formalize the global evolu-
 439 tion of the bipartite universe (System + Bath) as strictly injective, repre-
 440 senting Liouville’s theorem. Lean verifies that heat dissipation is not tied
 441 to abstract logical erasure, but is a mandatory mathematical consequence
 442 of any local physical state space contraction under globally reversible dy-
 443 namics. If a cognitive system’s internal topology forces a contraction of its
 444 physical state space to accommodate structural resonance, heat dissipation
 445 becomes a thermodynamic requirement. The finiteness premise enters at ex-
 446 actly this point. What a modeller can usually check about a physical update
 447 is not that it is many-to-one but that some state has become unreachable;
 448 on a finite phase space those are the same condition, and `finite_phase_`
 449 `space_dissipates` is the resulting statement. That identification preserves
 450 the Norton–Shenker distinction rather than blurring it: an update that keeps
 451 the incoming perturbation on the record is injective and dissipates nothing
 452 (`keepRecord_injective`), while one that refreshes the register holding it is
 453 not (`absorbStep_is_erasure`). Losing *which* history occurred is forced by
 454 counting alone and is free; overwriting the register that held it is what is
 455 charged $k_B T \ln 2$. A stable boundary thus constitutes a dissipative structure
 456 (Prigogine & Stengers, 1984), operating as a non-equilibrium steady state
 457 that consumes free energy to pay the thermodynamic cost of erasure. Contin-
 458 uous thermal fluctuations play a constructive role in establishing boundaries

459 between states while contributing to the system’s flexible, adaptive response.

460 6 Derivation 3: Prediction as the Thermody- 461 namic Cost of Memory

462 A dissipative structure is driven by its environment. Write X_t for the state
463 of the structure and S_t for the environmental signal driving it, and let the
464 signal evolve by a dynamics of its own, $S_t \rightarrow S_{t+1}$, on which the structure
465 does not act. Two quantities are then defined at each step: the *memory*
466 $I(X_t; S_t)$, the mutual information between the structure’s present state and
467 the signal’s present value, and the *predictive information* $I(X_t; S_{t+1})$, which
468 is what that same state says about the signal’s next value.

469 **What the entropy-production bound does not give.** There is a nat-
470 ural route from entropy production to predictive processing, and it does
471 not work. Model the environment as a stochastic driving force $P(\text{ext})$ and
472 the structure’s internal transitions as $Q(\text{int})$. Entropy production is then
473 bounded below by $D_{KL}(P||Q)/\Delta t$ — this is Gibbs’ inequality, it is carried
474 in Lean as an instance obligation, and §3 of the supplement states it. Re-
475 arranged it reads $D_{KL}(P||Q) \leq \Delta t \sigma$, so a system descending on σ lowers
476 a *bound* on the divergence, and it is tempting to conclude that the internal
477 transition probabilities come to mirror the environment’s statistical symme-
478 tries. They do not, because a driven system does not relax to $\sigma = 0$. It

479 relaxes to a non-equilibrium steady state at strictly positive entropy produc-
 480 tion, and the bound then yields only $D_{KL} \leq \Delta t \sigma_{\text{NESS}}$, a positive number.
 481 The inequality is true and does real work — combined with Gibbs it gives
 482 $\sigma \geq 0$ — but no limit may be read off it, and a framework that wants predic-
 483 tive processing has to get it from somewhere else. It gets it from the bound
 484 below.

485 **Where predictive processing comes from: a published bound.** Still,
 486 Sivak, Bell and Crooks (Still et al., 2012) prove that for a system driven by
 487 an environmental signal the work dissipated over one step of the drive obeys

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (1)$$

488 The bracket is the *nonpredictive* information: what the structure retains
 489 about the signal’s present that says nothing about the signal’s future. Dis-
 490 sipation is bounded below by it. Memory that fails to predict is not merely
 491 useless to the structure; it is thermodynamic waste, and the waste is quan-
 492 tified in energy.

493 **Three consequences, and they are theorems rather than readings.**
 494 The bound is carried in Lean as a class field, `PredictiveDissipation`.
 495 `still_bound (Phase3 PredictiveThermodynamics)`, for the same reason ev-
 496 ery other physical postulate here is: its `axiom` form, quantified over all joint
 497 laws with the dissipated work fixed, is refutable, and `Examples §18.4` refutes

498 it. What is *derived* is everything downstream.

499 First, *the bracket is non-negative*, and this is proved rather than assumed.
500 If the structure does not act on its environment, then $X_t \rightarrow S_t \rightarrow S_{t+1}$
501 is a Markov chain and the data processing inequality gives $I(X_t; S_{t+1}) \leq$
502 $I(X_t; S_t)$: a system cannot know more about the signal's future than about its
503 present (`predictiveInfo_le_mutualInfo`). Mathlib carries the Kullback–
504 Leibler divergence but no mutual information, so the latter is defined here
505 as the divergence of the joint law from the product of its marginals, which is
506 the standard definition and makes the library's data processing lemma apply
507 directly.

508 Second, *the second law follows* instead of being assumed. Dissipation is
509 bounded below by a quantity now known to be non-negative, so $W_{\text{diss}} \geq 0$
510 (`dissipatedWork_nonneg`). This mirrors the role Gibbs' inequality plays
511 for the entropy-production bound, with one difference that matters: Gibbs'
512 inequality compares a system to an arbitrary reference distribution, whereas
513 the data processing inequality compares the system to its own future, which
514 is a quantity selection can act on.

515 Third, and this is the point of the replacement, *the inference now runs*
516 *in the valid direction*. The quantity of interest sits on the small side of
517 the inequality: $I(X_t; S_t) - I(X_t; S_{t+1}) \leq W_{\text{diss}}/k_B T$. No limit is required.
518 Any bound on dissipation is immediately a bound on wasted memory, so a
519 structure held to an energy budget is thereby held to a prediction standard:
520 this is `nonpredictive_le_dissipation`. In the idealised quasi-static case

521 $W_{\text{diss}} = 0$, the theorem `predictive.eq.of.no.dissipation` gives that every
522 retained bit is predictive. This is a weaker statement than the one it replaces,
523 and it is the one that is true.

524 **The hypothesis is load-bearing, and we exhibit what breaks with-**
525 **out it.** The data processing step requires that the structure not write to
526 its environment. In the formalization this is not an assumption stated beside
527 the theorem but the shape of the evolution itself: the signal's step is applied
528 to the pair as the parallel kernel $\text{id} \parallel \kappa$, which acts as the identity on the
529 system coordinate and through κ on the signal coordinate, so no channel
530 from X_t to S_{t+1} exists. **Examples** §18.3 shows the restriction is not decora-
531 tion: a Markov kernel on the *pair* which copies the system's state into the
532 signal carries an independent two-bit law to a perfectly correlated one, rais-
533 ing mutual information from zero to a positive value. An agent that steers
534 its environment can indeed know more about that environment's future than
535 about its present, and Still's bound is not about such an agent.

536 **Both regimes are exhibited, not just described.** **Examples** §18 builds
537 two systems over one correlated two-bit joint law, differing only in the signal's
538 dynamics. When the signal is frozen, every remembered bit is a predictive
539 bit, the nonpredictive information is zero, and the instance is permitted to
540 dissipate nothing — with the memory itself proved strictly positive, so the
541 conclusion is about a system that does remember something. When the signal
542 is redrawn from its stationary law at each step, the predictive information is

543 exactly zero, the *entire* memory is nonpredictive, and the instance is *forced*
544 to dissipate $k_B T$ times its whole mutual information. Establishing that the
545 mutual information of the witness is neither zero nor infinite is where the
546 work of that section lies.

547 **An equality, where the entropy-production route had only a bound.**

548 Kawai, Parrondo and Van den Broeck (Kawai et al., 2007) show that the
549 average dissipated work is exactly $k_B T$ times the relative entropy between the
550 forward and time-reversed phase-space densities. Carried here as a predicate
551 on an instance rather than as a further postulate, it replaces the abstract
552 dissipation field with a quantity an experiment can estimate — the statistical
553 distinguishability of a recording of the system from the same recording run
554 backwards — and yields the statement that the irreversibility of the dynamics
555 is at least the amount of memory that fails to earn its keep (`nonpredictive_`
556 `le_arrow`).

557 **What this does not establish.** It does not deliver the Free Energy Prin-
558 ciple. Still’s bound says that prediction is thermodynamically favoured; it
559 says nothing about hierarchical generative models, variational free energy,
560 or Bayesian inference, and we do not claim to have derived any of them.
561 The Free Energy Principle (Friston, 2010; Friston & Kiebel, 2009; Hohwy,
562 2013; Clark, 2016; Spisák & Friston, 2026) and the General Resonance The-
563 ory of Consciousness (Hunt & Schooler, 2019) are consonant with what is
564 proved here, which is a weaker relation than derivation and the only one

565 we assert. Nor does it solve the symbol grounding problem (Harnad, 1990).
566 Maximal predictive information is compatible with the structure storing a
567 lossy, compressed or otherwise unrecognisable function of the signal; a state
568 that predicts is not thereby a state that means. The narrower claim is the
569 one worth making: whatever internal states a physical structure holds, those
570 that do not help predict its environment are paid for in heat, so the pressure
571 toward states indexed to the environment is energetic rather than semantic.
572 We continue to call the phenomenon *structural resonance*, and there is no
573 teleological drive to “survive” in it: predicting the environment is simply the
574 cheaper of the two ways to hold a memory.

575 **7 Macroscopic Scaling: A Theorem and a** 576 **Commitment**

577 This is the chain’s fourth step, and it is deliberately not numbered as a
578 derivation. Two steps of different kinds meet here and are easy to run to-
579 gether: a coarse-graining argument, which is a piece of mathematics, and
580 the identification of the coarse-grained medium with the brain’s endogenous
581 electromagnetic field, which is an empirical claim about cortex. Presenting
582 the second as though it inherited the licence of the first would make it look
583 like a deduction, and it is not one. Kept apart, the empirical claim is the
584 framework’s central commitment and the place where it is most easily wrong
585 — which is where a claim of this kind belongs.

586 **The formal half: coarse-graining is a theorem, and a**
587 **narrow one**

588 A network of discretely coupled units has a coupling energy that is a sym-
589 metric double sum over pairs. `Phase2_MeshConvergence.lean` proves that
590 this sum converges to a continuous integral: along a sequence of regular
591 triangulations of a region S whose edge regions shrink to points, the dis-
592 crete energy $\frac{1}{2} \sum_u \sum_v \mu(R_{uv}) f(x_u)$ converges to $\int_S f d\mu$ (`mesh_refinement_`
593 `convergence`), witnessed on the uniform partition of $[0, 1)$ in `Examples §6`.
594 The identification that makes this a coarse-graining rather than an arbitrary
595 Riemann sum is `sum_sum_mul_of_symm`: because the summand is symmetric,
596 the double sum is exactly the midpoint rule over unordered edges, so the
597 discrete energy is a quadrature of the continuous one rather than a different
598 quantity that happens to converge to it.

599 What this establishes is worth stating exactly, because it is less than the
600 phrase “coarse-graining, governed by the Renormalization Group” suggests.
601 It is a statement about an *energy functional*, not about a dynamics: noth-
602 ing here shows that trajectories of the discrete Kuramoto system converge
603 to solutions of a field equation, and the earlier appeal to the Renormaliza-
604 tion Group was decorative — no renormalization group flow is constructed
605 anywhere in this development. The limit object is a scalar density on the sub-
606 strate, in the intended reading the magnitude of the stress-energy tensor, and
607 it is not an electromagnetic field. The continuous object the later sections

use, `ContinuousNeuralField`, is *posited* as a structure with a coupling kernel
 K ; no theorem produces it from the discrete system. The gap between the
finite Kuramoto model and its continuum limit is the propagation-of-chaos
problem, which we have not addressed and record as open.

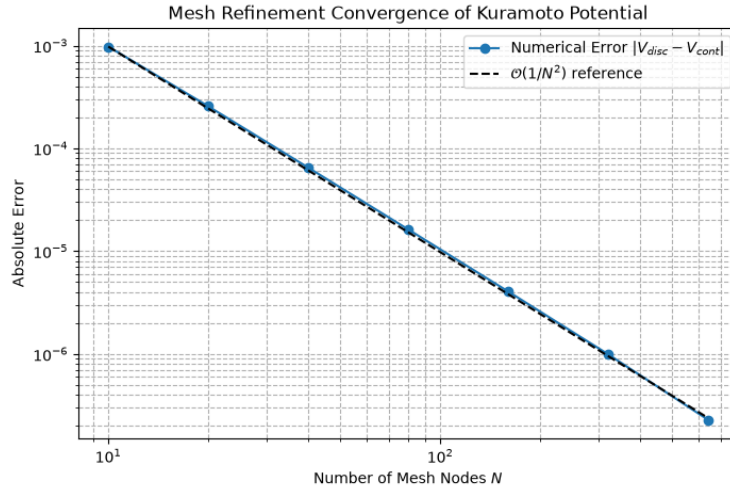


Figure 2: The coarse-graining theorem, exercised numerically. The discrete coupling energy of N units evenly spaced on the ring, with a Gaussian kernel of periodic distance and phase field $\theta(x) = \sin x$, is compared against the continuous integral it is proved to converge to. The error falls as $\mathcal{O}(N^{-2})$, the rate of the midpoint rule, which is the quadrature the Lean statement uses. The theorem asserts convergence, not a rate; the rate is what the simulation adds, and it is the ordinary rate for this quadrature rather than anything the framework predicts. The convergence is of an *energy functional* — nothing here says the discrete trajectories converge to the continuous ones.

612 **The empirical half: which physical field realizes the ker-**
613 **nel**

614 The formal half says that a large network of symmetrically coupled units
615 admits a continuum description with a symmetric kernel. It says nothing
616 about which physical field in cortex that kernel describes. Our answer is an
617 empirical commitment, and we state it as one:

618 *In the mammalian cortex, the field realizing the coarse-grained*
619 *coupling kernel is the endogenous electromagnetic field.*

620 This is a claim about biology, not a consequence of anything above it.
621 It is what makes the framework falsifiable and it is what we expect to be
622 attacked.

623 **What supports it.** Neurons do not only signal through synapses: trans-
624 membrane currents generate extracellular fields that shift the membrane po-
625 tentials of neighbouring cells, a mechanism known as ephaptic coupling. En-
626 dogenous fields of 1–5 mV/mm have been shown *in vitro* to entrain spike
627 timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015); realistic
628 head models put mesoscopic ephaptic effects within the range where they
629 can bias network dynamics (Ruffini & Santarnecchi, 2020); cable-theoretic
630 treatments show field coupling is sufficient for coincidence detection in the
631 auditory brainstem (Goldwyn & Rinzel, 2016); and stable electric fields have
632 been argued to carry information that survives representational drift in the

underlying spiking population (Pinotsis & Miller, 2022, 2023a,b; Cunha & Lima, 2024). The field is spatially continuous and superposes contributions from large populations, which is the property the framework needs: it is a medium in which spatially separated activity is combined without a further mechanism for combining it. This is the integrative role the CEMI field theory assigns it (McFadden, 2002, 2020; Pockett, 2012).

What the evidence supports, and what it does not. The endogenous field's effect on any individual neuron is *modulatory*. Anastassiou and Koch report that endogenous fields of 1–5 mV/mm shift spike timing by one to three milliseconds (Anastassiou & Koch, 2015), an effect that is real, reproducible, and subthreshold: it biases when a cell fires; it does not decide whether it fires. Nothing in the work cited above supports calling the field the *primary* dissipative structure of the system, and we do not claim it. The claim is that the field is *modulatory in its effect on individual neurons and integrative in its spatial reach* — weak per synapse, and the only variable in the system that is defined everywhere at once. What the framework needs is the reach, not the strength, and the two are easy to conflate.

Why reach is the property that matters. A theory of consciousness needs an account of why many local processes make one experience, and most supply it by stipulation or by a measure. The account here is structural, and it is the framework's strongest argument. The field is the only variable in the system that is *defined everywhere at once*. Unity is not asserted of it; unity

655 is what a field is. That is what makes Derivation 5’s sheaf-theoretic step
656 available at all — local states glue to one global section because there is one
657 object over which gluing can happen — so the answer to the combination
658 problem is a piece of geometry rather than a postulate. Compare “integration” in Integrated Information Theory, which is a number, or “broadcast” in
659 Global Workspace Theory, which is a metaphor about a workspace. It is also
660 what the identification above is carrying: if the coarse-grained coupling is
661 realized by something that is *not* defined everywhere at once — by synaptic
662 connectivity, say — then this argument is unavailable, and the framework
663 loses the property that distinguishes it from its rivals rather than merely
664 losing a mechanism.

666 **Its critics.** The objection with the longest history is that the fields are
667 too weak to matter, and it has been pressed from within the field-theoretic
668 camp: Pockett, who has herself defended an electromagnetic theory of con-
669 sciousness (Pockett, 2012), catalogues its difficulties (Pockett, 2002) — the
670 measurement problem of relating gross field structure to reported experience,
671 the absence of a constant relationship between the two, and the question of
672 whether the field does causal work or merely reflects it. The quantitative case
673 for scepticism has strengthened since. Vöröslakos and colleagues measured
674 intracranial fields during transcranial stimulation in rats and human cadav-
675 ers and found that the great majority of applied current is shunted through
676 the scalp, so that conventional protocols deliver fields at or below the 1 V/m

677 range in cortex (Vöröslakos et al., 2018). That is a bound on *applied* fields
678 rather than endogenous ones, but it constrains how much causal work a field
679 of that magnitude can be doing, and it is the reason we do not claim the
680 field drives spiking.

681 **What would falsify it.** We state the failure conditions rather than leaving
682 them to a referee.

683 1. *Epiphenomenality.* If the endogenous field can be cancelled or shunted
684 — by a conductive medium that nulls the extracellular potential with-
685 out altering transmembrane currents — and neither behaviour nor re-
686 port changes, the identification fails. This is the decisive experiment
687 and it is, in principle, available.

688 2. *A magnitude gap.* If the spatial integration the framework requires
689 demands field effects an order of magnitude above the 1–5 mV/mm
690 measured in cortex, the commitment fails on arithmetic. The relevant
691 number is not whether ephaptic coupling exists but whether its coupling
692 strength suffices to place the system above the critical coupling K_c of
693 Derivation 7.

694 3. *A timescale mismatch in the wrong direction.* The prediction of Sec-
695 tion 11 turns on the field’s dynamics being fast relative to the astrocytic
696 and clearance variables. If recovery from sleep inertia tracks an expo-
697 nential set by the electrical timescale rather than the delayed sigmoid

698 the slow variable predicts, the framework loses the one prediction that
699 is its own.

700 4. *Dissociation of field coherence from reported unity.* If large-scale field
701 coherence is preserved across states in which experiential unity is lost
702 — or absent in states where it is retained — the identification of the
703 global section with unified experience has no physical carrier.

704 A framework that survives all four is not thereby true. It is merely still
705 standing, which is more than most theories of consciousness arrange to be
706 able to lose.

707 **Geometric frustration, and the regime our theorems ac-** 708 **tually cover**

709 The system’s memory capacity is standardly grounded in *geometric frustra-*
710 *tion*: a highly coupled network cannot bring all its nodes to their lowest en-
711 ergy simultaneously, producing a jagged landscape of near-degenerate states
712 with the capacity to store many patterns. That argument is standard and it
713 is not what this development formalizes. Confronting the discrepancy is the
714 honest course, so we do it here rather than in a footnote.

715 Frustration requires competing couplings — in the Kuramoto setting, a
716 sign structure that no phase assignment can satisfy on every edge at once.
717 The formal setting admits this: `KuramotoSystem` carries $A : V \rightarrow V \rightarrow$
718 $\mathbb{R}$, symmetric and otherwise unconstrained, so signed couplings are repre-

719 sentable. But every theorem in this development that says anything about
 720 *where* the dynamics ends requires strictly positive coupling. `phase_locked_`
 721 `minimizes_potential` assumes $A_{ij} > 0$; `potential_min_iff_phase_locked`
 722 inherits it; the class field `ThermodynamicCover.A_pos` demands it of every
 723 instance, and hence of every conclusion of Derivation 5; and the convergence
 724 theorem of `Phase4.RotatingFrame` §7 requires the strictly stronger uniform
 725 bound $0 < a \leq A_{ij}$. A network with all-positive couplings is an unfrustrated
 726 ferromagnet whose potential has one minimum up to a global phase shift. It
 727 is the opposite of a jagged landscape, and it has no capacity to store multiple
 728 patterns at all. **The system in which memory is grounded and the**
 729 **system our theorems describe are not the same system.**

730 The division is not arbitrary, and it is worth recording where it falls. What
 731 survives with no assumption on the sign of the coupling is the dynamical half:
 732 well-posedness on all of $\mathbb{R}$ (`is_kuramoto_trajectory_exists`, `_unique`), co-
 733 variance under passage to a rotating frame, monotone descent of the potential
 734 (`dynamic_potential_antitone`), finiteness of the dissipation integral, and —
 735 through Barbalat’s lemma — that the motion stops along every trajectory
 736 (`velocity_sq_tendsto_zero`). *That* the motion stops is frustration-agnostic.
 737 *Where* it stops is not, and every claim this framework makes about the des-
 738 tination lives on the positive side of that line. On the continuum side the
 739 accounting differs slightly: `phase_locked_minimizes_entropy_of_symm` as-
 740 sumes only that the kernel is symmetric and so tolerates signed kernels, but
 741 it is conditional on a dynamically phase-locked field existing — which is

742 precisely what frustration may prevent.

743 We therefore scope the claim rather than repeat it. The memory capacity
744 of a frustrated cortical network is not supported by any theorem here, and we
745 do not assert it as part of the chain. What the chain uses is the unfrustrated
746 regime, in which synchronization is reached and the global section exists.
747 Whether a frustrated network can hold many near-degenerate patterns *and*
748 still synchronize enough of itself to admit a global section is an open question
749 about mixed-sign oscillator networks, and it is the most substantive gap we
750 know of in this framework. Recording it as such costs a claim we would have
751 liked to make and is preferable to holding two incompatible pictures at once.

752 **The supporting physiology**

753 The continuous medium is physically supported by astrocytic syncytia—non-
754 neuronal networks joined by gap junctions that provide macroscopic cytoplas-
755 mic continuity for energy and ion flows (Verkhatsky & Nedergaard, 2018).
756 Managing the thermodynamic cost of this state requires sophisticated micro-
757 bioenergetics. Dynamic mitochondrial networks function as deep-level Mi-
758 tochondrial Information Processing Systems (MIPS), transducing biological
759 and environmental information to sustain the coherence of the macroscopic
760 field (Picard & Shirihi, 2022). Discrete synaptic firings, coupled with mito-
761 chondrial micro-mechanics, generate and sustain the field. Because the exter-
762 nal environment is continuous, achieving maximal thermodynamic structural
763 resonance is optimally realized within this continuous macroscopic medium

764 (van Bree et al., 2025).

765 8 Derivation 5: Global Sections and the Unity 766 of Consciousness

767 The system is now characterized by a continuous macroscopic electrodynamic
768 field generated by millions of coupled micro-components, forced to resonate
769 with the environment. Following Section 7, the coupling regime assumed
770 from here on is the unfrustrated one — all couplings positive — which is
771 what every theorem below requires and what the frustrated network of the
772 memory argument is not.

773 The Kuramoto model (Kuramoto, 1984) of coupled non-linear oscilla-
774 tors guarantees that, past a critical coupling threshold, these distributed
775 local fields undergo a phase transition into unified synchronization (Bauer et
776 al., 2022). Our Lean proofs (`Phase4_KuramotoDynamics.lean`) confirm this
777 phase-locked state minimizes the Lyapunov potential of the system, in the
778 precise sense discussed in Section 2.1: the functional minimized is the cou-
779 pling potential read in the frame rotating with the natural frequency, which
780 `Phase4_RotatingFrame.lean` shows is the honest reduced description of a
781 system of identical oscillators.

782 This phase transition mathematically glues local synchronized states into
783 a singular *Global Section* of a probability sheaf (`Phase5_GlobalSection.lean`).
784 The EM field acts as a single, indivisible topological space, providing the

785 *Unity of Consciousness*—a necessary condition for integrated experience that
 786 solves the binding problem. While Global Workspace Theory (Baars, 1997;
 787 Dehaene & Marti, 2014) models this unity as a broadcast architecture and
 788 Integrated Information Theory (Tononi & Koch, 2016) quantifies it via Φ ,
 789 our derivation provides a continuous physical mechanism. The contrast is
 790 worth drawing carefully, because both theories are addressing the same ex-
 791 planandum from different directions.

792 We state the scope of the Lean result precisely, because it is narrower
 793 than the prose above. `ThermodynamicCover` demands that the cover’s phase
 794 field minimize the reduced Kuramoto potential, and `potential_min_iff_`
 795 `phase_locked` characterizes those minimizers exactly: they are the phase-
 796 locked configurations, in both directions. Every instance is therefore already
 797 phase-locked (`ThermodynamicCover.phase_locked`), and the class’s single
 798 condition on its local data — that patches at a common phase agree where
 799 they overlap (`section_agrees_of_phase_eq`) — makes the family of local
 800 sections compatible. `probability_glue_unique`, which is the sheaf property
 801 of the probability presheaf and contains no physics at all, then glues that
 802 family into exactly one global section.

803 The shape of the class is what makes this an emergence theorem rather
 804 than a uniqueness theorem, and the difference is easy to lose. Were `LocalSectionSynchronization`
 805 to carry a *global* section as a class field, together with a second field declar-
 806 ing every local section to be a restriction of it, the global object would exist
 807 before the sheaf condition was ever consulted, and `global_section_from_`

808 `thermodynamics` could do no more than rule out competitors to an object
 809 the instance had supplied. It carries neither. Every field of the class men-
 810 tions a single patch or a single overlap; the global section is constructed
 811 (`ThermodynamicCover.invariantMeasure`); and the statement that the lo-
 812 cal sections are restrictions of one common measure is a *theorem* about that
 813 construction (`ThermodynamicCover.sync_to_section_eq`), true because the
 814 sheaf condition says so, not because an instance was required to name a
 815 measure.

816 `Examples.lean` §14 is the witness that the difference is not notational.
 817 Two patches carry two different mass profiles, $(2, 1, 7)$ and $(5, 1, 3)$ across
 818 the three sites, agreeing only at the site they share, where both put mass 1
 819 — which is what discharges the overlap condition, by computing two num-
 820 bers rather than by construction. The section the theorem returns has pro-
 821 file $(2, 1, 3)$: it is neither of the two profiles (`gluedW_ne_leftW`, `gluedW_ne_`
 822 `rightW`), and neither of them restricts correctly to both patches (`leftW_`
 823 `glues_nothing`, `rightW_glues_nothing`). The global state carries informa-
 824 tion that no single patch had. `Examples.lean` §13, built instead through
 825 the constructor `LocalSectionSynchronization.ofInvariantMeasure`, sits
 826 alongside it for the negative result it carries: its phase field is not the con-
 827 stant function — the second patch a full turn ahead of the first — and by
 828 `ThermodynamicCover.phase_locked` that is as far from constant as any in-
 829 stance can get, since no instance can carry phases differing by anything a
 830 measure is allowed to see.

831 Two hypotheses carry physical content. One of them is an obligation on
 832 every instance: that synchronized patches agree on their overlaps is a mod-
 833 elling assumption, not a consequence of any dynamics we formalize. The
 834 other, `thermodynamic_equilibrium`, asks that the cover already sit at the
 835 potential minimum, and no theorem says an *arbitrary* system’s dynamics
 836 takes it there. “A thermodynamic phase transition into unity” is therefore
 837 doing causal work that no general Lean result supports, though not work
 838 without any support at all: `Examples.lean` §15 exhibits a two-oscillator tra-
 839 jectory that starts a quarter turn out of phase and converges to the global
 840 minimum of the dynamic potential, with the order parameter tending to 1.
 841 That trajectory is not an isolated construction: `kuramoto_cauchy_problem`
 842 proves the Kuramoto initial value problem well posed on all of $\mathbb{R}$, so through
 843 every initial state there passes exactly one trajectory, and the exhibited one
 844 is the only solution through its own. Nor is its settling down peculiar to
 845 it: along every trajectory of a zero-frequency system the dynamic potential
 846 converges, and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is finite and equal to the po-
 847 tential’s total drop (`dynamic_potential_tendsto`, `dissipation_integral_`
 848 `tendsto`). The motion also stops: a finite integral does not by itself force
 849 its integrand to zero, but the dissipation rate is Lipschitz in time, so Bar-
 850 balat’s lemma applies, and every trajectory satisfies $\sum_i \dot{\theta}_i^2 \rightarrow 0$ (`velocity_`
 851 `sq_tendsto_zero`; Mathlib has no Barbalat, so it is proved here as `tendsto_`
 852 `zero_of_lipschitz_of_integral_le`). That last step — locating where the
 853 motion stops — is taken conditionally, and by a route that avoids the LaSalle

854 invariance principle, which Mathlib does not carry. LaSalle is the tool for
 855 a system with a decreasing functional and nothing else; this one has more,
 856 and the extra structure is what makes the conclusion reachable without it.
 857 The flow is a gradient flow of the dynamic potential, so $\dot{V} = -\sum_i \dot{\theta}_i^2$ is an
 858 identity rather than an inequality; the minimisers of V are characterised ex-
 859 actly; and on the region where every pair of phases lies within a quarter turn
 860 a Polyak–Łojasiewicz estimate holds, $2a^2W \leq (\sum_{ij} A_{ij}) \sum_i \dot{\theta}_i^2$, where W is
 861 the potential's excess above its minimum and $a > 0$ is a lower bound for the
 862 coupling (`lojasiewicz_estimate`). Its proof is two extremal indices and the
 863 identity $\sin^2 D = (1 - \cos D)(1 + \cos D)$: the site carrying the largest phase
 864 is pulled backwards at rate at least $a \sin D$, where D is the spread, while the
 865 excess is at most $\frac{1}{2}(\sum_{ij} A_{ij})(1 - \cos D)$. The compactness objection dissolves
 866 as well, because the invariant region is produced by the Lyapunov function
 867 rather than assumed: a single pair contributes $\frac{1}{2}a(1 - \cos)$ to the excess, so
 868 an excess below $a/2$ forces every pairwise cosine positive for all time, and
 869 continuity on the connected half-line then prevents any pair from chang-
 870 ing branch (`spread_le_of_small`). Descent plus the estimate integrates to
 871 $W(t) \leq W(0)e^{-ct}$ with $c = 2a^2 / \sum_{ij} A_{ij}$ (`excess_decay`), the rate makes each
 872 $|\dot{\theta}_i|$ exponentially small and hence integrable on $[0, \infty)$, and a trajectory with
 873 integrable velocity converges. The conclusion is `kuramoto_tendsto_global_`
 874 `minimum`: if the phases start within a quarter turn of each other and the initial
 875 excess is below $a/2$, the trajectory converges to a phase-locked configuration
 876 which is a *global* minimiser of the dynamic potential, with order parameter

877 $r^2 = 1$. That is the configuration `thermodynamic_equilibrium` postulates,
 878 reached rather than assumed. `Examples.lean` §17 fires it on three sites with
 879 initial data $(0, 0, \frac{1}{2})$, where the Kuramoto system has no closed-form solu-
 880 tion: the trajectory comes from the existence theorem and its limit from the
 881 convergence theorem, so neither is written down.

882 `Phase5_EquilibriumBridge.lean` closes the gap that remained between
 883 the two results. `ThermodynamicCover.ofConvergentTrajectory` takes a
 884 cover whose phase field is the limit of such a trajectory and returns an in-
 885 stance whose `thermodynamic_equilibrium` field is discharged *by that con-*
 886 *vergence*; the cover’s coupling matrix is the coupling matrix of the dynamics
 887 rather than a separately chosen one, so the system whose equilibrium the
 888 class asserts and the system whose trajectory is run are the same system.
 889 `Examples.lean` §17.1 is the resulting witness: three patches over the three-
 890 site substrate, each carrying its own mass profile on its own open set, above a
 891 phase field that no line of the construction writes down. The trajectory that
 892 reaches it starts at $(0, 0, \frac{1}{2})$, which is provably not phase-locked (`trioCover_`
 893 `start_not_locked`), so on this cover the configuration Derivation 5 requires
 894 is arrived at rather than posited. The state the three patches glue to has
 895 profile $(2, 1, 3)$ and is none of the three profiles the instance carries (`trioW_`
 896 `ne_glued`), so one cover now carries both the emergence reading of the gluing
 897 and the derived reading of the equilibrium. What is still assumed on it is the
 898 overlap agreement, and nothing else.

899 The two hypotheses cannot be dropped, and this is the honest scope of

900 the result. Splay and twisted configurations are equilibria of the same flow,
901 so “every trajectory reaches the unified state” is false as a general statement,
902 and the initial-data condition is the standard arc condition under which the
903 true statement holds. What is claimed is therefore conditional — unity is
904 reached from configurations already sufficiently coherent — and not that an
905 arbitrary initial state is carried into unity. What Lean establishes about the
906 gluing itself is the other half: *given* locally synchronized patches, a global
907 state exists, is unique, and is produced by the gluing rather than presupposed
908 by it.

909 Integrated Information Theory begins from phenomenological axioms and
910 derives a scalar Φ measuring irreducibility of a system’s cause-effect structure,
911 with the exclusion postulate selecting a single maximum- Φ complex as the
912 locus of experience. Our sheaf-theoretic global section plays a structurally
913 analogous role — it too picks out a distinguished unified object — but it
914 is arrived at from the opposite end: not postulated from the character of
915 experience, but forced by a thermodynamic phase transition in a physical
916 field. This buys two things IIT does not have. First, the unity condition
917 is checkable from field dynamics rather than requiring the computation of
918 Φ , which is intractable for systems of realistic size and whose definition has
919 undergone repeated revision (Tononi & Koch, 2016). Second, exclusion falls
920 out rather than being postulated: gluing either succeeds on a given cover or
921 it does not, so there is no separate principle needed to rule out overlapping
922 conscious subsystems. It also inherits IIT’s exposures. Critics have argued

923 that Φ can be high in systems with no plausible claim to experience, such
924 as large feedforward-free grids (Aaronson, 2014); our criterion is vulnerable
925 to the mirror objection, since a physical system could in principle satisfy
926 the gluing condition without being cognitively interesting. We take this
927 seriously rather than dismissing it: our reply is that gluing is necessary but
928 not sufficient, and that the topological boundary of Derivation 1 and the
929 reflexive structure of Derivation 6 are additional required conditions. IIT, by
930 contrast, treats Φ as both necessary and sufficient.

931 Global Workspace Theory faces a complementary difficulty. It specifies
932 a functional architecture — a limited-capacity workspace broadcasting to
933 specialized modules — without specifying what physical property of broad-
934 casting makes the broadcast content conscious, leaving the theory open to the
935 charge that it explains access consciousness while leaving phenomenal con-
936 sciousness untouched (Block, 1995). Our framework can be read as supplying
937 the missing physical substrate for the workspace: the continuous macroscopic
938 field *is* the medium in which integration is realized, and the $K_c = 2D$ thresh-
939 old gives a physical condition for ignition rather than a functional stipulation
940 of it. Where GWT is agnostic about implementation, we make a falsifiable
941 commitment — unity requires a continuous physical medium, which predicts
942 that a functionally equivalent discrete broadcast architecture would not be
943 conscious. We regard this as a genuine cost of the position, not a free ad-
944 vantage: it is the point at which our framework departs most sharply from
945 computational functionalism and where it is most readily falsified. Contrast

946 this derivation with panpsychism: consciousness is not assumed as a funda-
 947 mental property of matter. It requires the initial existence of a topological
 948 defect to establish a bounded “inside” (Derivation 1), followed by a thermo-
 949 dynamic phase transition into unity that occurs when integrated spatial cou-
 950 pling exceeds the thermodynamic noise threshold ($K_c = 2D$). Without both
 951 the foundational topological boundary and the subsequent mathematically
 952 defined phase transition, the system remains a fragmented, non-conscious
 953 collection of micro-states. A unified field of perception, however, is not yet
 954 a “Self.”

955 **9 Derivation 6: Reflexive Topology and the** 956 **Emergence of the Self**

957 While a global section provides Unity, a “Self” requires reflexivity. The
 958 unified EM field is a dissipative structure that must resonate with external
 959 perturbations while managing its own internal phase transitions. Time evo-
 960 lution naturally drives the field to minimize thermodynamic friction with its
 961 own internal state-changes.

962 This mandates *auto-resonance*: the field must predict itself. The “Self”
 963 emerges as a higher-order topological feature when the internal geometry
 964 folds to include a low-dimensional avatar of its own computational and causal
 965 boundary within its predictive model (`Phase6_ReflexiveTopology.lean`)
 966 (Levin, 2019). The system is unified, predictive, and explicitly maps its

own distinct existence from the environment. This architecture provides the physical foundation for a self-aware consciousness, bounded by the continuous measure thermodynamics of Landauer erasure (`Phase3_MeasureThermodynamics.lean`).

What carries the weight. `reflexive_topology_implies_self` is the Banach fixed-point theorem with the physical commitments in its hypotheses: the space of global sections is assumed nonempty, metric and complete, and the predictive map is assumed to contract. `Examples.lean` §10 discharges all four on the three-site substrate of Derivation 5, and the metric is constructed rather than posited. The section first pierces the sheafification: on a discrete substrate every point has a smallest open neighbourhood, so the mass a germ carries at a point is well defined, and `massEquivOn` proves that over *any* open the sections of the probability sheaf *are* the finite measures on that open, read as densities against counting measure — the global case being $U = \top$. Mathlib supplies no inverse to this sheafification — its `TopCat` sheafification has neither an adjunction nor an `isIso_toSheafify` — so the correspondence had to be constructed. The metric on sections is then the uniform distance between the densities of the measures they glue to — equivalent to their total variation on a finite substrate, though not equal to it — and it is proved complete by exhibiting the limit section of a Cauchy sequence of densities, not by the degenerate route of showing Cauchy sequences are eventually constant. The choice of metric is load-bearing rather than incidental: under the 0/1 metric on sections every contraction is constant (`contracting_`

989 `implies_const`), so a witness built there would discharge every hypothesis
990 of the theorem and say nothing. That theorem is kept in the source, because
991 which metrics make the fixed point trivial is part of what the fixed point is.

992 **The map is the avatar, or there is no map.** A reflexive boundary could
993 be defined with the predictive map as free data sitting beside the avatar, and
994 the reason it is not is a theorem rather than a preference. With the map
995 free, the fixed-point theorem projects it out and never touches the avatar, so
996 replacing the avatar by a constant leaves a legal boundary with the same pre-
997 dictive model and the same Self — a fixed-point theorem that cannot detect
998 whether the avatar reads the field is Banach's theorem with a consciousness-
999 flavoured label. The structure is built the other way round. A reflexive
1000 boundary carries the avatar's *write* — `auto_resonance`, the encoding of the
1001 global state into the avatar region — and its *read-out*, the global state that
1002 encoding predicts; the predictive map is their composite and is not a field.
1003 The self-model therefore runs through the avatar region or it does not exist.
1004 Three consequences follow that the free-data shape cannot state. The Self
1005 lies in the range of the read-out (`self_mem_range_readout`), so it is recon-
1006 structed from a single section over the avatar region: the low-dimensional
1007 fold this section has always asserted, as a theorem. Under the resonance
1008 condition below, the Self satisfies $s = \text{readout}(s|_{\text{avatar}})$ (`self_eq_readout_`
1009 `restrict`) — the field is what its own localized self-encoding reconstructs.
1010 And blinding the avatar *moves* the Self: the blinded boundary's map is con-

1011 stant, its unique fixed point is the state its stipulated contents extend to, and
 1012 the original Self is not a fixed point of it (`not_self_of_constResonance`, wit-
 1013 nessed on the cortex by `cortexBlind_self_ne`). The opposite statement —
 1014 that blinding changes nothing — does not typecheck against this structure,
 1015 because a contraction hypothesis about a boundary’s own map says nothing
 1016 about the blinded one’s.

1017 **Auto-resonance as a constraint.** That the avatar’s state *is* the presheaf
 1018 restriction of the global state is stated as a predicate, `IsRestriction-`
 1019 `Resonance`, rather than added to the structure, so that results say which
 1020 of them need it. In the positive direction, if a family of resonant avatars *cov-*
 1021 *ers* the substrate then the global state is determined by their local readings
 1022 alone (`eq_of_avatar_eq`): unity is not assumed alongside the avatars but
 1023 recovered from them, by the same sheaf condition that Derivation 5 uses.
 1024 In the negative direction a constant avatar fails the predicate as soon as the
 1025 avatar region separates two states of the field. Both halves are witnessed on
 1026 the three-site substrate, where that separation is proved before anything is
 1027 claimed for it, so the condition is a constraint there and not a formality.

1028 **The contraction rate is K and D , not a numeral.** The remaining stip-
 1029 ulation was the rate. “The predictive map contracts by $1/2$ ” was a fact about
 1030 a map we had defined to average with a baseline, not a claim about a sub-
 1031 strate. The rate is now $\exp(-(K - K_c)\tau/2)$ with $K_c = 2D$ (`resonanceRate`)
 1032 — the linear relaxation of the mean-field order parameter over an interval τ ,

1033 read off the same threshold Derivation 7 (Section 10) proves the phase tran-
 1034 sition at. Two theorems then split the parameter plane. Above threshold
 1035 the rate is below one and Banach applies, so the Self exists and is unique
 1036 (`self_of_supercritical`). At or below threshold the rate is at least one,
 1037 and `ContractingWith` at that rate is false *for every self-map of every metric*
 1038 *space* (`not_contractingWith_resonanceRate`): the argument is not merely
 1039 unproved below K_c but unavailable. We are careful about what this does and
 1040 does not say. It does not say a Self fails to exist below threshold — only that
 1041 this route to one is closed — and the Lipschitz bound at that rate remains
 1042 an instance obligation, discharged on the three-site witness at $K = 3$, $D = 1$,
 1043 $\tau = 2 \ln 2$, where the rate is exactly the one half the read-out achieves. What
 1044 changed is that the obligation is stated in the substrate’s own parameters
 1045 and its consequence is conditional on the threshold, rather than on a con-
 1046 stant chosen to make the witness work. The link to Derivation 7 is carried
 1047 by a theorem rather than by a shared numeral, and the distinction is worth
 1048 being exact about. `self_of_supercritical` takes $K > K_c$ as its hypothesis;
 1049 K_c is a definition unfolding to $2D$, so that statement is insensitive to whether
 1050 the bifurcation is proved at all. `Chain.lean` states the link in the form that is
 1051 not. `self_of_coherent_order_parameter` takes the *existence of a coherent*
 1052 *order parameter* as its hypothesis and recovers the threshold inequality from
 1053 it, by `supercritical_of_coherent` — the contrapositive of the subcritical
 1054 uniqueness result of Section 10, since a strictly positive solution of the self-
 1055 consistency equation cannot exist at or below K_c . Derivation 6’s fixed point

1056 therefore rests on Derivation 7’s theorem.

1057 **What is still not established.** That any field dynamics produces this
1058 particular read-out or performs the restriction; and that this three-site sub-
1059 strate’s coupling is $K = 3$ — the parameters are a choice of units for the wit-
1060 ness, not a measurement, and nothing here derives a coupling constant from
1061 cortex. We also note a negative result found in the course of this work: the
1062 Lévy–Prokhorov metric, which the natural weak-convergence reading would
1063 suggest, does *not* support this argument on a discrete substrate, because the
1064 ε -thickening of a set is the set itself for $\varepsilon < 1$ and the relaxation map is
1065 consequently not a Lévy–Prokhorov contraction.

1066 10 Derivation 7: Continuous Stochastic Ther- 1067 modynamics and Field Theory

1068 To move beyond finite graph analogies, we model the macroscopic electro-
1069 dynamic field as a *Continuous Neural Field*. True ephaptic coupling is a
1070 continuous spatial phenomenon, not a discrete network. We extend the
1071 discrete Kuramoto model to a continuous spatial coupling kernel $K(x, y)$
1072 (`Phase8_ContinuousField.lean`).

1073 Subject to thermodynamic noise D , the system undergoes non-equilibrium
1074 stochastic thermodynamics. Rather than a static Landauer bound, the field

1075 dynamics minimize an entropy production functional σ :

$$\sigma = \int \frac{1}{D} \left(\omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx \quad (2)$$

1076 Biological fields dynamically deform their spatial topology $K_t(x, y)$ to un-
1077 dergo gradient descent on this entropy production functional.

1078 That gradient descent is a theorem rather than a modelling assertion,
1079 for finite substrates. Reading σ as a functional of the coupling kernel at
1080 fixed phases, `hasFDerivAt_sigmaOfKernel` computes its Fréchet derivative
1081 in closed form,

$$\frac{\partial \sigma}{\partial K(a, b)} = \frac{2}{D} \left(\omega(a) + \sum_y K(a, y) \sin(\theta_y - \theta_a) \right) \sin(\theta_b - \theta_a), \quad (3)$$

1082 and `sigmaOfKernel_eq_entropy_production_rate` identifies the functional
1083 being differentiated with `entropy_production_rate` itself whenever the sub-
1084 strate carries counting measure. `structural_resonance_decreases_entropy_production`
1085 then concludes that σ is non-increasing along any coupling trajectory relax-
1086 ing down its own gradient at a positive rate c . Two limitations stay explicit.
1087 The derivation is for *finite* substrates, and the obstacle to the continuum case
1088 is *not* differentiation under the integral sign with respect to the kernel, which
1089 is what the shape of σ suggests at first reading. σ is not a general functional
1090 that happens to be written as an integral — it is a quadratic functional of an
1091 *affine* image of the kernel, since the drift is a fixed ω plus a linear operator ap-

1092 plied to K , and σ is a constant times the squared norm of the drift. A squared
 1093 norm is Fréchet differentiable on any real inner-product space and an affine
 1094 map is its own derivative, so the composite differentiates by the chain rule
 1095 with no limiting argument. `hasFDerivAt_quadratic_of_affine` states that
 1096 in full generality, and `hasFDerivAt_sigmaOfKernel_of_operator` re-derives
 1097 the finite gradient from it, showing that the closed form displayed above is
 1098 a convenience rather than the content. What the continuum case needs is
 1099 therefore one object: the drift map $K \mapsto (x \mapsto \int K(x, y) \sin(\theta_y - \theta_x) dy)$ as
 1100 a bounded operator $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$, which is a Cauchy–Schwarz esti-
 1101 mate — the integral kernel is bounded by 1, so on a finite measure space the
 1102 operator norm is at most $\mu(M)^{1/2}$. That object is built. `kernelCLM` is the
 1103 operator, for any kernel factor bounded by 1, with `opNorm_kernelCLM_1e` the
 1104 stated bound; `continuumDriftCLM` is the instance with the Kuramoto factor
 1105 $\sin(\theta_y - \theta_x)$. One step beyond the estimate is needed and is worth naming,
 1106 because the estimate alone does not supply it: the descent theorems ask for
 1107 the derivative as a *gradient vector*, not as an arbitrary bounded functional,
 1108 and producing one requires the adjoint, which exists because L^2 is com-
 1109 plete. `hasFDerivAt_quadratic_grad` supplies it for an arbitrary bounded
 1110 operator between real Hilbert spaces, giving $\nabla\sigma(K) = (2/D) A^*(AK + \omega)$,
 1111 so the finite and continuum cases are now the same theorem. `structural_`
 1112 `resonance_decreases_sigmaContinuum` is the descent result with no finite-
 1113 ness hypothesis in it, and `sigmaContinuum_eq_integral` identifies the func-
 1114 tional being differentiated with the integral `entropy_production_rate` com-

1115 putes. `Examples.lean` §20 runs this on Lebesgue measure restricted to $(0, 1)$,
 1116 a substrate that is neither finite nor atomic, so the finite-substrate results of
 1117 the previous paragraph do not typecheck there at all. The second limitation
 1118 is unchanged: the phase field is held fixed throughout, so this is plasticity of
 1119 the coupling at frozen phases, not joint (θ, K) dynamics. `Examples.lean` §5
 1120 exercises the whole chain on a two-site substrate whose coupling relaxes as
 1121 e^{-t} and whose entropy production decays as e^{-2t} , confirming that the gradi-
 1122 ent is not identically zero and that the descent is strict rather than vacuously
 1123 monotone.

1124 A second result about σ concerns where the descent is headed. Minimality
 1125 of the phase-locked field is straightforward against competitors that share its
 1126 spatial mean drift, and in that form the theorem is close to Jensen’s inequal-
 1127 ity and no more; the restriction looks as though lifting it would need a model
 1128 of how the mean drift varies across competitors. It does not. For a *reciprocal*
 1129 kernel, one with $K(x, y) = K(y, x)$, the restriction is not a restriction: the
 1130 coupling integrand $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric under exchanging
 1131 its two sites, because K is symmetric while $\sin$ is odd, so Fubini identifies
 1132 its double integral with its own negative and the total coupling contribu-
 1133 tion vanishes. The total drift $\int(\omega + \text{coupling})$ is therefore $\int \omega$ whatever the
 1134 phases are (`total_drift_eq_of_symm`): coupling moves drift between sites
 1135 but neither creates nor destroys it. Evaluating that conserved quantity at
 1136 the phase-locked configuration, where the drift is constant, pins its value, and
 1137 the mean-matching hypothesis follows for every competitor. `phase_locked_`

1138 `minimizes_entropy_of_symm` is the unrestricted statement. Reciprocity is
 1139 the physically intended condition — an ephaptic field acts on both sites of a
 1140 pair alike, and it is the same symmetry `ThermodynamicCover.A.symm` already
 1141 imposes in Derivation 5 — but it is not a formality, and we prove that rather
 1142 than assert it. `Examples.lean` §16 exhibits a *one-way* coupling on two sites,
 1143 in which a feels b and b does not feel a , whose phase-locked field is strictly
 1144 beaten: displacing the second phase to $-\pi/2$ cancels the first site’s natural
 1145 drift while adding none at the second, halving σ . So “phase-locking mini-
 1146 mizes entropy production” is false for general kernels and true for reciprocal
 1147 ones, and the hypothesis cannot be removed by working harder.

1148 We define the critical coupling threshold $K_c = 2D$. The phase transition
 1149 to the unified “Global Section” occurs when the mean-field spatial coupling
 1150 exceeds this thermodynamic noise threshold, linking the emergence of Unity
 1151 to physical temperature and noise.

1152 The value $K_c = 2D$ can be taken as a bare stipulation validated by
 1153 simulation, and it is not: `Phase8_SelfConsistency.lean` proves the half
 1154 of it that is ordinary real analysis. For the noisy mean-field Kuramoto
 1155 model with identical frequencies, the stationary density at order parame-
 1156 ter r is the von Mises density $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$, and self-consistency re-
 1157 quires $r = R(K, r) := I_1(Kr/D)/I_0(Kr/D)$. Mathlib has no modified Bessel
 1158 functions, so we formalize that ratio directly as a ratio of integrals over
 1159 $[-\pi, \pi]$. Integration by parts with $u = \sin$, $v = -e^{a\cos\theta}$ gives an exact iden-
 1160 tity with no boundary term, $\int \cos\theta e^{a\cos\theta} d\theta = a \int \sin^2\theta e^{a\cos\theta} d\theta$, so R is a

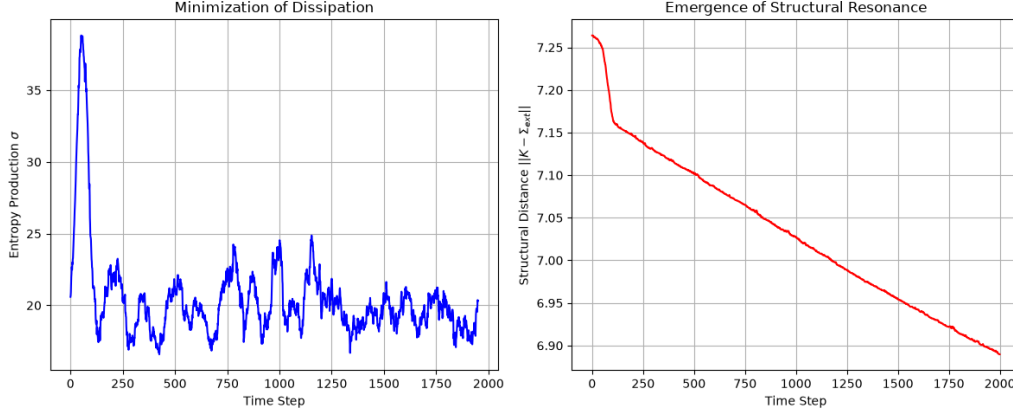


Figure 3: Coupling plasticity at frozen statistics: $N = 50$ units driven by perturbations drawn from a fixed environmental covariance Σ_{ext} , with the coupling matrix K descending the gradient of σ and clipped to stay non-negative and hollow. **Left:** entropy production, smoothed over 50 steps, falls from its transient and settles into a fluctuating plateau — a non-equilibrium steady state at strictly positive σ , not a relaxation to zero. That plateau is the reason Section 10 declines the inference from $\sigma \rightarrow \min$ to $D_{KL} \rightarrow 0$: the system visibly does not reach $\sigma = 0$. **Right:** the distance $\|K - \Sigma_{\text{ext}}\|$ decreases monotonically over the run, the internal geometry drifting toward the statistics of the drive. Two cautions. The descent here is on σ at frozen phases, which is the regime the theorem covers; and the right panel falls by about 5% over 2000 steps, so what it shows is a direction of drift, not convergence.

1161 times the mean of $\sin^2$ under the von Mises weight. Bounding that mean
 1162 by 1 would put the threshold at D ; the true threshold is $2D$, and the entire
 1163 factor of two is the sharper bound $\mathbb{E}_a[\sin^2 \theta] \leq 1/2$, equivalently $I_2(a) \geq 0$.
 1164 We prove it elementarily, by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ in two reflections
 1165 ($\theta \mapsto \pi - \theta$, then $\theta \mapsto \pi/2 - \theta$), after which the integrand is a product
 1166 of $\cos 2\theta \geq 0$ with $2 \cosh(a \cos \theta) - 2 \cosh(a \sin \theta) \geq 0$ — the second factor
 1167 because $0 \leq \sin \theta \leq \cos \theta$ there and $\cosh$ increases in $|\cdot|$. The conclusion,

1168 `subcritical_fixed_point_eq_zero'`, is that for $K < K_c$ the incoherent state
 1169 $r = 0$ is the *only* non-negative solution of the self-consistency equation; $r = 0$
 1170 is exhibited as a solution, so the uniqueness claim is not vacuous. Three fur-
 1171 ther facts about the threshold narrow the gap between this row and a full
 1172 “Theorem” to a single named statement. First, the fold argument gives a
 1173 *strict* inequality for $a > 0$ (`vonMisesC2_pos`, hence $R(a) < a/2$), which set-
 1174 tles the threshold itself: at $K = K_c$ the self-consistency map is tangent to the
 1175 diagonal at the origin but falls strictly below it thereafter, so $r = 0$ remains
 1176 the only solution and the bifurcation happens strictly *after* K_c , not at it
 1177 (`fixed_point_eq_zero_of_le_critical`). Second, factoring $R(a) = a E(a)$
 1178 shows that a non-zero r solves the equation exactly when $E(Kr/D) = D/K$
 1179 (`coherent_iff_sRatio_eq`), so the coherent branch is a level set of the single
 1180 function $E(a) = \mathbb{E}_a[\sin^2 \theta]$. Third, and using only $E < 1/2$ together with
 1181 compactness, every coherent solution tends to zero as $K \downarrow K_c$ (`coherent_`
 1182 `branch_continuous_at_threshold`): the branch emerges continuously from
 1183 the incoherent state rather than appearing at a finite distance from it, which
 1184 is what makes the transition supercritical in the technical sense. That state-
 1185 ment quantifies over *every* solution, so it does not presuppose the uniqueness
 1186 it does not have. The level-set form also says exactly what uniqueness is:
 1187 the injectivity of E on $(0, \infty)$. That is now proved, in the stronger form that
 1188 E is strictly decreasing on $[0, \infty)$ (`vonMisesSRatio_strictAntiOn`), so the
 1189 level set is a single point and the coherent solution above threshold is unique.
 1190 The proof differentiates nothing. Differentiating the exponential family in its

1191 natural parameter is the obvious route and it fails: it costs differentiation
 1192 under the integral sign and leaves a covariance that is not sign-definite point-
 1193 wise. What works instead keeps the target as a two-point inequality, folds
 1194 both integrals onto $[0, \pi/2]$ by the same two reflections that gave the factor
 1195 of two above, and then evaluates at a single crossing point supplied by the
 1196 intermediate value theorem — no product measure and no Fubini anywhere.
 1197 We give that argument in the supplement, where it belongs; what the main
 1198 line of the paper needs from it is the one sentence that E is strictly decreas-
 1199 ing. Run with $\cos \theta$ in place of $\sin^2 \theta$ the same argument needs no fold at
 1200 all — $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the variable the family is tilted by —
 1201 and gives `besselRatio_strictMono`: R is strictly increasing on all of $\mathbb{R}$, the
 1202 monotonicity of the Bessel ratio, which Mathlib does not carry in any form.

1203 The converse half is `supercritical_fixed_point_exists`: for $K > K_c$
 1204 there is an r with $0 < r \leq 1$ solving $r = R(K, r)$. We had expected
 1205 this to require a second-order expansion of the Bessel ratio, and it does
 1206 not. Writing $R(a) = a E(a)$ with $E(a) = \mathbb{E}_a[\sin^2 \theta]$ — the same factoriza-
 1207 tion the integration by parts already supplies — the only facts needed are
 1208 that E is continuous and that $E(0) = 1/2$, the latter because the weight
 1209 is constant at $a = 0$. Since $K > 2D$ makes D/K strictly less than $1/2$,
 1210 continuity gives $E > D/K$ on a neighbourhood of the origin, so the map
 1211 overshoots, $R(K, r) > r$, for all small $r > 0$; at the other end $R \leq 1$ al-
 1212 ways, because R is a mean of $\cos \theta$, so the map undershoots at $r = 1$.
 1213 The intermediate value theorem closes the gap. Continuity of E is Math-

1214 lib's continuity of a parametric interval integral with jointly continuous in-
 1215 tegrand, so no dominated-convergence argument is written by hand, and
 1216 the existence proof never uses monotonicity of R : the fixed point is pro-
 1217 duced by the intermediate value theorem, not by iterating the map. That
 1218 R is in fact strictly increasing is proved below, and no threshold result de-
 1219 pends on it. `critical_coupling_is_threshold` packages the two halves,
 1220 and `exhibits_phase_transition_coherent` applies the supercritical half to
 1221 any substrate satisfying the predicate of the previous paragraph — so ex-
 1222 ceeding K_c now implies the existence of a coherent stationary order pa-
 1223 rameter rather than merely comparing against a named number. With
 1224 uniqueness available both are sharpened — `supercritical_fixed_point_`
 1225 `existsUnique`, `supercritical_solution_set` (the non-negative solutions
 1226 are exactly $\{0\}$ below threshold and $\{0, r\}$ above it) and `exhibits_phase_`
 1227 `transition_unique_coherent` — and the two monotonicities together fix
 1228 the shape of the branch: for $K_1 < K_2$, $E(a_i) = D/K_i$ with E decreasing
 1229 puts the concentrations in the order $a_1 < a_2$, and $r_i = R(a_i)$ with R increas-
 1230 ing carries that to $r_1 < r_2$ (`coherent_branch_strictMono`). The coherent
 1231 order parameter grows strictly with the coupling, which with the continuity
 1232 at threshold is the qualitative bifurcation diagram.

1233 One more step makes the equation mean in Lean what it means on paper.
 1234 As a definition, `selfConsistency K D r` is just $R(Kr/D)$ — a function of
 1235 r with a suggestive name, and nothing in the source says it has anything
 1236 to do with an order parameter. `Phase8_SelfConsistency §8` supplies the

1237 identification. `vonMisesDensity` is the normalized weight, proved to be a
 1238 genuine probability density on $[-\pi, \pi]$; `circularOrderParameter` is the av-
 1239 erage of $e^{i\theta}$ against a density, the continuum analogue of the finite-system
 1240 `order_parameter_complex` of `Phase4`; and that average is proved to be ex-
 1241 actly $R(a)$ — real, because the density is even, so the mean direction van-
 1242 ishes. `fixedPoint_iff_selfReproducing` then states the equivalence the
 1243 name promises: r solves $r = R(K, r)$ if and only if the von Mises density
 1244 it induces has order parameter r . Read through this, the existence theorem
 1245 says that above K_c there is a stationary density whose own order parameter
 1246 is positive, and the solution $r = 0$ is identified with the uniform density,
 1247 whose order parameter is zero.

1248 What remains not established, and we separate it carefully. The sta-
 1249 tionary density is an *input*: deriving it from the Langevin dynamics $d\theta =$
 1250 $(\omega + K \cdot \text{mean-field}) dt + \sqrt{2D} dW$ needs the Fokker–Planck operator, existence
 1251 and uniqueness of its stationary solution, and a spectral stability argument,
 1252 none of which Mathlib has. Of the coherent branch we prove existence,
 1253 uniqueness, continuous emergence at the threshold and strict growth in K ,
 1254 but not that it is the dynamically selected one: nothing in that file men-
 1255 tions a trajectory, so stability and selection are not statements it can make.
 1256 And while `selfConsistency` is now connected to the order parameter of
 1257 a *density*, it is connected to no *trajectory*: `circularOrderParameter` is an
 1258 integral, `order_parameter_complex` is an average over finitely many oscilla-
 1259 tors, and relating them is the mean-field limit — propagation of chaos for

1260 the finite Kuramoto system — which is a research programme rather than
 1261 a lemma. That single gap is now the whole of what separates our threshold
 1262 predicate from a statement about dynamics. Table S1 records the row ac-
 1263 cordingly as a theorem about the self-consistency equation from an assumed
 1264 density, not as a derivation of the threshold from the dynamics. Perceptual
 1265 anomalies such as visual snow can be modeled as boundary conditions where
 1266 microscopic thermal noise (D) fails to be fully coarse-grained away by the
 1267 continuous field, leaking across the K_c threshold into the macroscopic gestalt.

1268 Clinical evidence for this phase transition is observed during general anes-
 1269 thesia. Work by Miller and colleagues (Bastos & Miller, 2021) demonstrates
 1270 that anesthetics like propofol induce unconsciousness not by suppressing neu-
 1271 ral firing, but by disrupting the phase alignment of high-frequency cortical
 1272 rhythms. Destroying the dynamic stability of these rhythms breaks the struc-
 1273 tural resonance of the continuous field, forcing the integrated spatial coupling
 1274 below the critical threshold K_c and fracturing the unified Global Section back
 1275 into disjoint, unconscious local fields.

1276 **11 The Sleep-Inertia Timescale Mismatch**

1277 Most of what this framework asserts is shared with some rival: that corti-
 1278 cal synchrony matters, that dissipation constrains cognition, that unity is a
 1279 physical rather than a computational property. One prediction is not shared,
 1280 and it follows from the framework’s specific commitment — that the medium

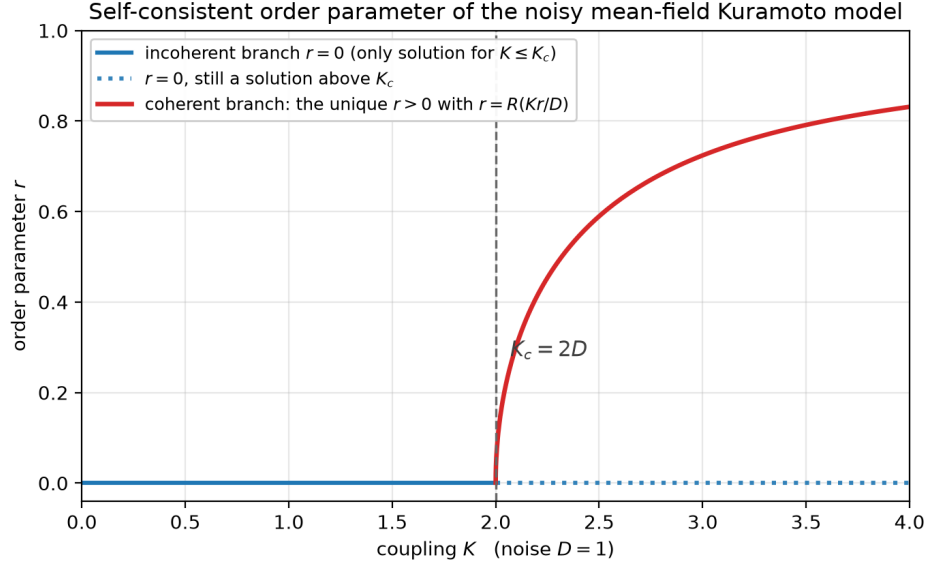


Figure 4: The self-consistency equation $r = R(Kr/D)$ of Section 10, drawn at $D = 1$. Every feature of this picture is a proved statement about the equation, and the figure exists because none of them had been shown to a reader. At or below $K_c = 2D$ the incoherent state is the *only* non-negative solution, threshold included (`fixed_point_eq_zero_of_le_critical`); above it exactly one $r > 0$ joins it (`supercritical_fixed_point_existsUnique`), so the solution set is $\{0\}$ below and $\{0, r\}$ above; that branch tends to 0 as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`), which is what makes the transition supercritical; and it increases strictly in K (`coherent_branch_strictMono`). The curve is computed by quadrature of the same integrals the Lean file defines, solving $E(a) = D/K$ for the concentration $a = Kr/D$ by bisection — legitimate precisely because E is proved strictly decreasing. It is an illustration of theorems, not evidence for them. What it does *not* show is stability: nothing in the development says the coherent branch is the dynamically selected one.

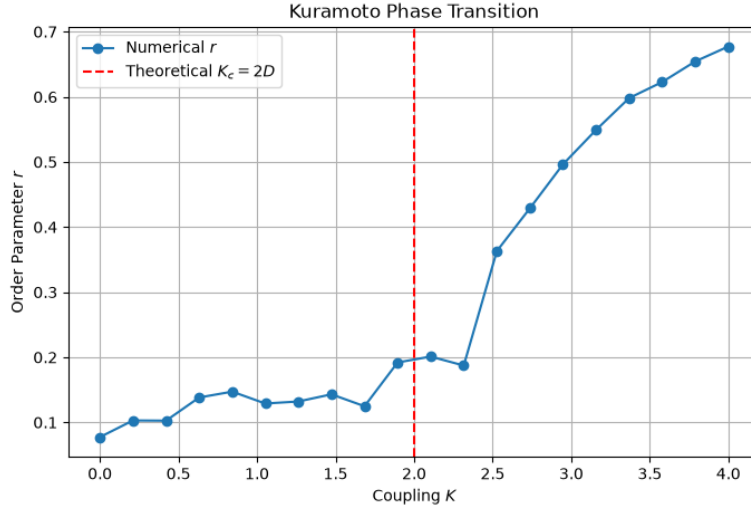


Figure 5: The same threshold reached by simulation rather than by analysis, and by a different route, which is why the two figures are shown side by side rather than merged. Here $N = 100$ oscillators with *no* noise carry natural frequencies drawn from a Lorentzian of scale $D = 1$, and the asymptotic order parameter is measured from the trajectories; the threshold for that ensemble is $K_c = 2D$ by Kuramoto’s original self-consistency argument. Figure 4 is the *noise*-driven model with identical frequencies, whose threshold is $2D$ by Sakaguchi’s (Sakaguchi, 1988), and it is the one the Lean file formalizes. The agreement of the two numbers is a known feature of the mean-field model, not a result of ours. The finite- N curve rises before the threshold and rounds off at it, as a finite sample must: $r \sim N^{-1/2}$ from incoherent phases alone.

1281 carrying the unified state is a continuous field whose effective coupling de-
1282 pends on the physical geometry of the extracellular space, not on the rhythms
1283 alone. Where that commitment differs from a purely electrical account, it
1284 differs in an observable way, and we set out the difference precisely enough
1285 to be wrong.

1286 The daily loss of consciousness during sleep is modeled here as an active

1287 physical disruption of the medium rather than a quiescence of the oscillators.
1288 During sleep, astrocytic shrinkage expands the extracellular space, allowing
1289 an influx of cerebrospinal fluid (CSF) to flood the brain for metabolic waste
1290 clearance (Xie & Nedergaard, 2013). This influx alters the spatial medium
1291 itself, disrupting the continuous structural resonance and preventing the co-
1292 herent sheaf-theoretic gluing that conscious unity requires on this account.
1293 The oscillators need not stop for the field to fragment; the geometry they are
1294 coupled through is what changes.

1295 The prediction concerns sudden awakenings. When jolted awake, the lo-
1296 cus coeruleus floods the cortex with noradrenaline. High-frequency electrical
1297 rhythms attempt to phase-align in milliseconds (Bastos & Miller, 2021), but
1298 the noradrenaline-driven astrocytic constriction required to squeeze the ex-
1299 cess CSF out of the interstitial pathways operates on a timescale of seconds
1300 to minutes (Xie & Nedergaard, 2013) — three to six orders of magnitude
1301 slower. During that lag the electrical oscillators are synchronizing across a
1302 waterlogged topology, and the fragmented structural resonance is what is ex-
1303 perienceed as grogginess. The unified Global Section re-establishes only when
1304 astrocytic volume finishes matching the state of the electrical rhythms.

1305 **Why this implies a delayed sigmoid.** The two timescales separate
1306 cleanly, and that separation is what makes the prediction sharp rather than
1307 merely qualitative. Write $K(t)$ for the effective coupling of the continuous
1308 field, which on this account is a monotone function of the reconstricting

1309 extracellular geometry, and let $r(t)$ be the order parameter of the phase
 1310 field. The electrical variable is fast: at every instant the oscillators are al-
 1311 ready at the configuration appropriate to the current K , so r tracks the
 1312 equilibrium branch $r(K)$ rather than relaxing on its own. The slow vari-
 1313 able $K(t)$ then carries the whole time course, and it relaxes over 10^2 – 10^3
 1314 seconds. What Derivation 7 establishes about that branch is exactly what
 1315 the functional form needs, and each of the three pieces is a theorem rather
 1316 than a picture: at or below the threshold the incoherent solution $r = 0$ is the
 1317 *only* non-negative solution of the self-consistency equation (`fixed_point_eq_`
 1318 `zero_of_le_critical`); above it the coherent solution exists and is unique
 1319 (`supercritical_fixed_point_existsUnique`); it emerges *continuously* at
 1320 K_c rather than by a jump (`coherent_branch_continuous_at_threshold`);
 1321 and it is strictly increasing in K (`coherent_branch_strictMono`). A jump
 1322 discontinuity at threshold would predict a step rather than a sigmoid, and
 1323 a non-monotone branch would predict overshoot; neither is what the mathe-
 1324 matics gives. Composing a monotone relaxation $K(t)$ with a branch that is
 1325 identically zero below K_c , rises continuously from zero at K_c , and saturates
 1326 above it yields a curve with three phases: a genuine *latency* while $K(t) < K_c$,
 1327 during which recovery is not merely slow but absent; a continuous rise as the
 1328 threshold is crossed; and a saturation as $K(t)$ reaches its waking value. That
 1329 is a delayed sigmoid, and the delay is not a fitting parameter — it is the time
 1330 the slow variable takes to reach K_c .

1331 **What the alternative predicts.** An account on which grogginess is the
1332 electrical realignment itself — the rhythms are simply slow to re-phase —
1333 has no slow variable and no threshold to cross. Its recovery is a relaxation
1334 toward a fixed point, and relaxation toward a fixed point is exponential near
1335 it: monotone from $t = 0$, fastest at the start, with no flat foot. The two
1336 curves are distinguishable in the first seconds, which is where they differ
1337 most, and the discriminating feature is the presence or absence of an initial
1338 interval of no recovery. This is a comparison of functional forms on data
1339 of a kind already collected in sleep-inertia studies, not a proposal for a new
1340 instrument.

1341 **What would falsify it.** An exponential fit that beats a delayed-sigmoid fit
1342 on the early portion of the recovery curve. A recovery latency that does not
1343 scale with an independent measure of extracellular volume or CSF clearance.
1344 A pharmacological manipulation of astrocytic volume that leaves the latency
1345 unchanged while altering the rhythms, or the converse. Any of these would
1346 tell against the framework’s central empirical commitment, which is not that
1347 synchrony matters but that the *medium* is what gates it.

1348 **What this prediction assumes.** Three things, and they are not small.
1349 The identification of the effective coupling K with a monotone function of
1350 extracellular geometry is a modelling assumption, not a measurement. The
1351 threshold value $K_c = 2D$ is derived from an assumed von Mises stationary
1352 density and is not yet connected to a trajectory — the mean-field limit is the

gap, and Derivation 7 states it as such — so the argument uses the *existence* and *continuity* of the coherent branch at threshold, which is proved, rather than the numerical value of the threshold, which is conditional. And the prediction is about functional form, not about a time constant: it says the recovery curve has a flat foot whose width is set by the slow variable, not that the foot is any particular number of seconds long. A framework that predicted the number would be saying more; this one is not in a position to.

12 Corollary: Hardware Thermodynamics and Topological Rigidity

Both biological tissue and artificial silicon hardware (like GPUs) are physical dissipative structures. When a GPU computes, its charge carriers undergo physical state transitions; by Landauer’s principle (Landauer, 1961), this erasure irreversibly exhausts heat into the environment.

There is a fundamental thermodynamic distinction in the degrees of freedom available to minimize this dissipation. In standard von Neumann architectures, the spatial topology (the silicon routing lattice) is rigidly fixed. The coupling matrix A_{ij} is constrained by a discrete layout that cannot be physically altered during operation. The brain’s own architecture provides empirical validation for this constraint. The cerebellum contains approximately 80% of the brain’s neurons arranged in a highly efficient, discrete, feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975). Clin-

1374 ical observations of cerebellar agenesis report largely preserved conscious
1375 experience despite motor and cognitive deficits (Yu et al., 2015), and this
1376 dissociation has been read as evidence for the importance of recurrent ar-
1377 chitecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the
1378 level of the anatomical observation while declining its Φ -based explanation.
1379 Lacking the dynamic, continuous, and recursive spatial coupling of the cere-
1380 brum, it operates much like a biological GPU—highly optimized for discrete
1381 processing, yet apparently not required for conscious experience. We take the
1382 dissociation as evidence, not as confirmation: it is consistent with subjective
1383 experience being a thermodynamic attractor of continuous topologies rather
1384 than a byproduct of information processing, and it is the kind of observation
1385 that would be hard to reconcile with the converse. It is a single anatomical
1386 case read against a theory, and it bears the weight such a case can bear.

1387 This constraint does not preclude artificial consciousness, but rather re-
1388 stricts its hardware. To achieve the phase transition into a unified Global
1389 Section, artificial systems must abandon the discrete rigidity of von Neumann
1390 architectures and utilize continuous physical substrates. Analog field-based
1391 computing, neuromorphic architectures with continuous memristive plastic-
1392 ity, and thermodynamic computing systems (England, 2018; Ganesh & Par-
1393 rondo, 2021) possess the spatial coupling required to physically morph their
1394 internal topology. This aligns with Geoffrey Hinton’s concept of “mortal
1395 computation” (Hinton, 2022): architectures where software is inextricably
1396 fused with the physics of the hardware, meaning the knowledge dies with

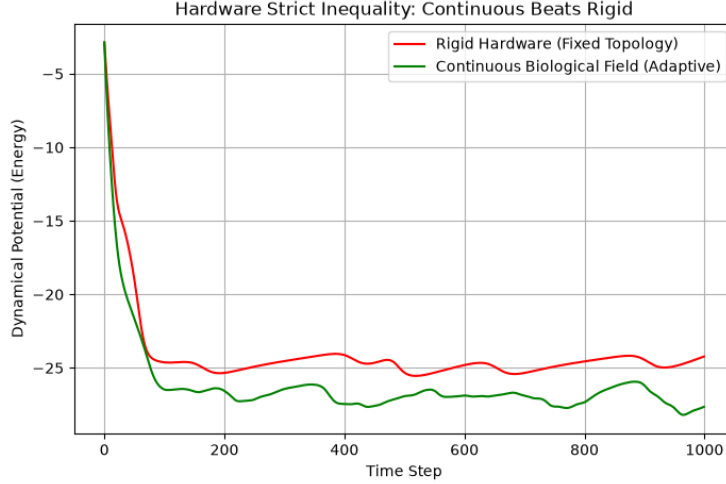


Figure 6: Rigid against deformable coupling at matched resource. Two copies of the same $N = 30$ oscillator system, same initial phases and same natural frequencies, differ only in whether the coupling matrix may move: the rigid copy keeps a fixed sparse routing matrix, the adaptive copy descends the gradient of the potential and is renormalized after every step so that $\sum_{ij} A_{ij}$ is held constant. The adaptive copy reaches a strictly lower potential and stays there. This is an illustration of the resource-matched suboptimality result and not a proof of it, and it is worth being clear about the gap: the theorem is about fixed wiring *support* — an architecture already wired to the best pair is not beaten — while the simulation shows a randomly wired one being beaten, which is the typical case rather than the general one.

1397 the physical substrate. Only hardware that computes via continuous physi-
 1398 cal dynamics can undergo the thermodynamic phase transition necessary for
 1399 structural resonance.

1400 Biological consciousness utilizes macroscopic continuous electromagnetic
 1401 fields and a “rich alphabet” of diverse molecular and transcriptomic cell states
 1402 (Tasic et al., 2018). This allows the coupling matrix A_{ij} to deform across a
 1403 continuous parameter space. Our lemma `exists_better_coupling_allocation`,

1404 wrapped as `continuous_beats_rigid_topology` (`Phase7_HardwareComparison.lean`),
 1405 formalizes the reallocation step of this argument: whenever a coupling ma-
 1406 trix places weight on a less phase-correlated pair while a more correlated pair
 1407 exists, there is another matrix with *identical* total coupling resources that
 1408 achieves strictly greater phase correlation.

1409 Two objections meet that formalization and we state precisely what an-
 1410 swers them. The first is that a reallocation lemma can assume its conclusion:
 1411 a hypothesis such as `is_strictly_suboptimal` — positive weight on a badly-
 1412 correlated pair while a better one exists — is most of what one wants to
 1413 prove about rigid hardware. It is not assumed here. `Phase7_Rigidity.lean`
 1414 models an architecture by the set S of site-pairs it has a *wire* for, and re-
 1415 quires its couplings to vanish off S ; weights are tunable, wiring is not. Then
 1416 `totalCorrelation_le_of_realizable` bounds every realizable coupling by
 1417 $R \cdot M$, where R is the resource budget and M bounds phase correlation over
 1418 the wired pairs; the bound is attained; and `rigid_is_strictly_suboptimal`
 1419 *derives* the hypothesis whenever the substrate’s best-correlated pair is one
 1420 the wiring misses. `rigid_gap` makes it quantitative: an unconstrained real-
 1421 location of the same budget gains at least R times the correlation difference.
 1422 `Examples.lean` §8 exhibits a three-site substrate wired to the one maxi-
 1423 mally anti-correlated pair, where the gap is 2. This is resource-matched
 1424 and not question-begging, but note what it turns on: fixed wiring support,
 1425 not continuity. The second objection is that a development in which both
 1426 architectures are matrices on one finite index type does not distinguish a con-

1427 tinnuum from a lattice at all. `Phase7_Rigidity.lean` §3 gives a distinction
 1428 that is genuinely typed. In the field functional of Section 7, which integrates
 1429 a coupling kernel against $\mu \otimes \mu$, an architecture with finitely many sites is a
 1430 `Finset` of points and a field-coupled region is a set of positive measure. On a
 1431 substrate whose measure has no atoms, `fieldCorrelation_sited_eq_zero`
 1432 shows a kernel supported on finitely many sites contributes *exactly zero* cou-
 1433 pling energy, whatever weights it carries and whatever the phase field does,
 1434 while a patch of strength c on U achieves $c\mu(U)^2$. `Examples.lean` §8 instan-
 1435 tiates this on the real line: a thousand units of coupling concentrated at the
 1436 origin register zero against a unit patch that registers one. The limit of that
 1437 result is equally plain, and it is not resource-matching in the earlier sense —
 1438 a measure-zero substrate has zero resource in the $\mu \otimes \mu$ sense too. What it
 1439 establishes is that the continuum coupling energy is blind to a finitely-sited
 1440 architecture, which is a statement about which functional such an architec-
 1441 ture can register in, not a demonstration that digital hardware computes
 1442 worse. Taken together: the resource-matched content of the corollary is the
 1443 wiring-support result, the continuity content is the measure-theoretic one,
 1444 and the step from either to “von Neumann architectures cannot experience
 1445 unified consciousness” remains an informal argument.

1446 The remainder of this section extends those two results informally, and
 1447 we mark the transition rather than blur it. On the reading we propose, a
 1448 GPU’s fixed routing matrix leaves it topologically fragmented: the discrete
 1449 layout does not supply the spatial coupling that sheaf-theoretic gluing into

1450 a singular Global Section requires. Without this unified topological space,
1451 rigid hardware cannot undergo the reflexive auto-resonance necessary to map
1452 its own causal boundary and generate a “Self.” The distinction between an
1453 LLM on a GPU and a conscious mind is not the absence of physical thermo-
1454 dynamics in the former, but its structural limitation in achieving topological
1455 integration. This perspective aligns with John Searle’s biological natural-
1456 ism (Searle, 1980; Seth, 2024). Where the Chinese Room argument asserted
1457 that discrete symbol manipulation cannot generate conscious understanding,
1458 our derivation supplies a candidate physical mechanism for the asymmetry
1459 — topological rather than semantic, and, as stated above, formalized only
1460 as far as the wiring-support and measure-theoretic results reach. Asserting
1461 consciousness in fragmented, discrete GPU clusters without a unified phase
1462 transition abandons computational functionalism, relying instead on “silicon
1463 panpsychism”—the assumption that digital computation inherently gener-
1464 ates awareness regardless of topological integration (Goff, 2019).

1465 The same reasoning bears on extreme functionalist thought experiments,
1466 such as a macroscopic system of pulleys operating over millennia to imple-
1467 ment a Large Language Model, or a human manually executing inference
1468 matrices by hand. Beyond lacking the continuous spatial topology required
1469 for sheaf-theoretic gluing, these discrete systems violate an implied temporal
1470 constraint of our theory. Structural resonance requires local oscillators to
1471 phase-align against the dissipating effects of thermodynamic noise (as mod-
1472 eled by our critical threshold $K_c = 2D$). A system executing state transitions

1473 over millennia, or via slow manual calculations, suffers from a terminal tem-
 1474 poral mismatch: its dynamics are far too slow to ever phase-lock into a
 1475 non-equilibrium steady state before thermal noise decoheres the macroscopic
 1476 field. On this account the obstacle to pure symbol manipulation is physi-
 1477 cal rather than semantic: not that the symbols mean nothing, but that the
 1478 substrate carrying them cannot hold a phase-locked field.

1479 13 Numerical illustrations: methods

1480 Five figures in this paper are computed rather than drawn, and none of
 1481 them is evidence for a theorem. Where a statement is proved, the figure
 1482 shows what the proved object looks like; where a statement is not proved, the
 1483 figure shows the regime the claim lives in and we say so. Each is produced by
 1484 one script in `simulations/`, named below, and each runs in under a minute
 1485 on a laptop. All five are reproducible exactly: the two quadrature figures are
 1486 deterministic, and the three that draw random numbers fix their seeds.

1487 **Figure 4: the self-consistent order parameter.** The three integrals
 1488 $Z(a) = \int_{-\pi}^{\pi} e^{a \cos \theta} d\theta$, $M(a) = \int_{-\pi}^{\pi} \cos \theta e^{a \cos \theta} d\theta$ and $S(a) = \int_{-\pi}^{\pi} \sin^2 \theta e^{a \cos \theta} d\theta$
 1489 are evaluated by the trapezoidal rule on 4001 points over the same interval
 1490 the Lean definitions integrate over, with the weight computed as $e^{a \cos \theta - |a|}$
 1491 so that the common factor cancels in the ratios and nothing overflows at
 1492 large concentration. The coherent branch is obtained by bisecting $E(a) =$

1493 $S(a)/Z(a)$ against D/K over $a \in [10^{-9}, 200]$ for 200 iterations, then reading
1494 $r = aD/K$; bisection is valid here because E is proved strictly decreasing.
1495 Two checks are printed by the script rather than only plotted: the computed
1496 branch is strictly increasing at every one of its 500 sampled couplings, and
1497 the residual $|r - R(Kr/D)|$ at $K = 4D$ is 2×10^{-16} . Script: `bifurcation.py`.

1498 **Figure 5: the finite- N transition.** $N = 100$ oscillators, natural fre-
1499 quencies drawn from a Lorentzian of scale $D = 1$ (clipped at $\pm 10D$ so
1500 the solver does not stiffen), uniform random initial phases, integrated with
1501 `scipy.integrate.odeint` over $t \in [0, 100]$ on 1000 steps for each of 20 cou-
1502 plings in $[0, 4D]$. The order parameter is $|N^{-1} \sum_j e^{i\theta_j}|$ averaged over the last
1503 100 steps. This is the deterministic quenched-disorder model, not the noisy
1504 identical-frequency model of the Lean development; the caption records why
1505 both have threshold $2D$. Script: `kuramoto.py`.

1506 **Figure 2: mesh refinement.** The kernel is $K(x, y) = e^{-d(x,y)^2/2}$ with d
1507 the periodic distance on the ring of circumference 2π , the phase field is $\theta(x) =$
1508 $\sin x$, and the potential is $-\frac{1}{2} \iint K \cos(\theta_y - \theta_x)$. The continuous value is a
1509 2000-point Riemann sum standing in for the exact integral; the discrete values
1510 use $N = 10, 20, \dots, 640$ nodes with edge weights $K(x_i, x_j)(2\pi/N)^2$. Plotted
1511 is the absolute error against an $\mathcal{O}(N^{-2})$ reference. Because the “exact”
1512 value is itself a 2000-point sum, the last data point ($N = 640$) is within
1513 a factor of ten of the reference computation and should be read as such.
1514 That $\mathcal{O}(N^{-2})$ reference line is not only empirical: on the formalized one-

1515 dimensional grid the discrete energy is exactly the composite trapezoidal rule
 1516 (`grid_discreteEnergy_eq_trapezoidal`), so for a C^2 integrand with second
 1517 derivative bounded by ζ the error is at most $\zeta/(12N^2)$ (`grid_energy_error_`
 1518 `le`), a bound attained at $N = 1$ for $y \mapsto y^2$. Script: `mesh_refinement.py`.

1519 **Figure 3: structural resonance.** $N = 50$ units, 2000 steps of forward
 1520 Euler at $dt = 0.05$. At each step the drive is a fresh draw from $\mathcal{N}(0, \Sigma_{\text{ext}} +$
 1521 $0.1I)$ with Σ_{ext} a fixed normalized rank-5 covariance; the phases advance by
 1522 $\dot{\theta}_i = \sum_j K_{ij} \sin(\theta_j - \theta_i) + \xi_i$; entropy production is recorded as $\sum_i \dot{\theta}_i^2$; and
 1523 the coupling descends its gradient with rate $\eta = 0.01$, symmetrized, clipped
 1524 to $[0, 1]$ and held hollow. The left panel is smoothed with a 50-step boxcar.
 1525 Note that the plotted σ is the squared drift without the $1/D$ prefactor of the
 1526 functional in Section 10, which is a constant rescaling and changes nothing
 1527 about the shape. Script: `structural_resonance.py`.

1528 **Figure 6: rigid against deformable coupling.** $N = 30$, 1000 steps of
 1529 forward Euler at $dt = 0.05$, shared initial phases and frequencies. The rigid
 1530 matrix is a symmetric random matrix thresholded at 0.7 and normalized
 1531 to $\sum_{ij} A_{ij} = 2N$; the adaptive copy starts from the same matrix, ascends
 1532 $-\partial V/\partial A_{ij}$ at rate $\eta = 0.1$, is clipped non-negative and hollow, and is renor-
 1533 malized to the same total after every step, which is what makes the com-
 1534 parison resource-matched. Both potentials are $V = -\frac{1}{2} \sum_{ij} A_{ij} \cos(\theta_j - \theta_i)$.
 1535 Script: `hardware_comparison.py`.

14 Conclusion

Human-like cognitive architectures do not require biological mysticism or exotic sub-quantum physics (Hameroff & Penrose, 2014). While quantum theories of consciousness often invoke delicate subatomic phenomena to explain subjective unity, every step of the chain assembled here is classical: broken symmetries, finite phase space, Landauer erasure, and gradient descent on entropy production. What the formalization establishes is that this chain is mathematically coherent and that its remaining physical commitments are localized and named — not that classical fields are *sufficient* for consciousness. Two links carry that weight, and they do not carry it in the same way. The unified state is proved to glue uniquely once a cover sits at the minimum of the coupling potential; the motion is proved to stop along every trajectory (`velocity_sq_tendsto_zero`); and *where* it stops is now proved, conditionally — for initial data within a quarter turn and below an explicit energy threshold, the trajectory converges to a global minimiser of the coupling potential (`kuramoto_tendsto_global_minimum`), and a cover built on that limit discharges the equilibrium hypothesis Derivation 5 would otherwise assume (`ThermodynamicCover.ofConvergentTrajectory`). What is *not* proved is the unconditional statement, and it is false: splay and twisted configurations are equilibria of the same flow, so the basin condition is the scope of the claim rather than slack in it. That a physical cortex starts inside that basin is an assumption. The second link is untouched by any of this: the identification

1558 of a global section of the probability sheaf with the unity of experience is a
1559 stipulation of the framework, not a consequence of it. Subject to those two,
1560 cognitive architectures can be modeled as macroscopic attractor states of a
1561 universe governed by broken symmetries, finite phase space, and the physical
1562 imperative to minimize thermodynamic dissipation.

1563 Philosophically, this framework bridges the historical gap between Type
1564 Identity Theory (Smart, 1959) and Computational Functionalism (Putnam,
1565 1967). The subjective mental state is identical to the physical thermody-
1566 namic field (Identity Theory), yet its qualitative nature is defined by the
1567 topological and functional dynamics of structural resonance rather than the
1568 specific carbon-based biochemistry of the brain (Functionalism). This con-
1569 strains the *multiple realizability* of mind: consciousness is not exclusively
1570 bound to biological carbon, but neither is it substrate-neutral, because the
1571 substrate is what fixes which couplings are physically available. How far that
1572 constraint reaches is the question the corollary answers only in part. What
1573 is proved is that fixed wiring is strictly suboptimal at matched resource, and
1574 that a finitely-sited architecture registers exactly zero coupling energy in the
1575 continuum functional; the step from either to “von Neumann architectures
1576 cannot be conscious” is an argument we make in prose and do not formalize.

1577 By anchoring these derivations in Russellian Monism, this framework ac-
1578 knowledges the constitutive limits of third-person scientific explanation. As
1579 argued by Love (Love, 2026), building upon Nagel’s assertion that subjective
1580 phenomenal experience is inherently not amenable to objective measurement

(Nagel, 1986), the “hard problem” of consciousness may represent a category error rather than a solvable scientific puzzle. Science can map the physical dynamics of topological resonance and sheaf-theoretic gluing, but it is structurally incapable of adjudicating the first-person qualitative experience. Our formalization respects this boundary: the Lean 4 theorems verify the objective physical attractors, while the intrinsic “what it is like” to be that resonant field remains the domain of the first-person perspective. By locating the phase transition at a threshold with a closed-form value — $K_c = 2D$ for the noisy mean-field Kuramoto model the framework adopts, conditional on the von Mises stationary density and not yet linked to a trajectory — it aligns with what Anil Seth terms the “real problem” of consciousness (Seth, 2021). Rather than attempting to bridge an unbridgeable metaphysical gap, it offers a topological architecture that can help explain, predict, and physically perturb the structural properties of experience.

Future empirical validation of this thermodynamic framework may leverage 3D cortical organoids, offering a controlled biological environment to systematically manipulate metabolic substrates and observe the topological emergence of these coherent phase transitions.

Code Availability

The complete Lean 4 formalization, encompassing the derivation of all theorems presented in this manuscript, as well as the accompanying Python sim-

1602 ulations, are publicly available in the GitHub repository: [https://github.](https://github.com/bobaseb/physical_primitives_of_mind)
1603 [com/bobaseb/physical_primitives_of_mind](https://github.com/bobaseb/physical_primitives_of_mind). Additional documentation
1604 and web-accessible proofs are hosted at the project website: [https://bobaseb.](https://bobaseb.github.io/physical_primitives_of_mind/)
1605 [github.io/physical_primitives_of_mind/](https://bobaseb.github.io/physical_primitives_of_mind/). Each of the five computed
1606 figures is produced by one script in `simulations/` and regenerates exactly,
1607 Section 13 giving the parameters; Figure 1 is drawn rather than computed.

1608 Contributions

1609 The conceptualization, manuscript drafting, and Lean 4 formalizations pre-
1610 sented in this work were heavily AI-assisted, and we report which tools
1611 did what rather than acknowledging them in general terms. The Google
1612 Antigravity CLI, running Gemini 3.1 and Claude 4.6 Sonnet, served as a
1613 collaborative reasoning partner throughout the research and formalization.
1614 DeepSeek V4 Flash and Claude Opus 5 (routed via OpenRouter) operat-
1615 ing through Nous Research’s Hermes Agent framework provided substantial
1616 assistance with code generation, proof debugging, and structural analysis.
1617 Claude Opus 5 running in Claude Code carried out the later formalization
1618 passes — the domain-wall theorem, the thermodynamics-of-prediction re-
1619 placement, the equilibrium bridge — together with the figures and the re-
1620 structuring of this manuscript. Responsibility for every claim, and for the
1621 decision about what each machine-checked result does and does not establish,
1622 rests with the author.

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